

SCHOOL OF COMPUTER SCIENCE

An Embodied Robotic Companion for Attention Support in ADHD

Luke Kim

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Abstract

Maintaining attention when sitting at a desk is a challenge for individuals with ADHD or subclinical attention difficulties. While body doubling is an effective strategy to combat attention difficulties, it depends on finding another person willing to body-double with them. Productivity tools available today rely upon screen-based notifications to remind the user of their pending tasks, but these notifications lack any social component.

In this dissertation, a robot is designed to autonomously detect when an individual with ADHD becomes distracted from their tasks and to respond to those distractions with non-verbal, anthropomorphic behaviours. The robot utilises the webcam attached to the user's laptop to detect their posture and whether they are distracted by another screen (such as their phone), through the MediaPipe BlazePose and YOLO libraries. When distraction is detected, the robot's lamp-inspired arm turns towards the user to provide a non-verbal cue to return to their tasks.

A formative survey of twelve participants — seven of whom self-identified as having ADHD — provided five requirements for the robot: to turn its gaze towards the user, to avoid any aversive or pressuring behaviours, to utilise a character-based design instead of a humanoid one, to communicate only through movement rather than sound or screen-based notifications, and to feel supportive rather than monitoring. The most clearly preferred behaviour was orientation-based gaze, with 9 of 12 participants rating it helpful or very helpful, directly shaping the system's primary intervention.

Finally, this dissertation contributes a working prototype of such a robot, a vision-based pipeline for distraction detection, an empirically derived set of design requirements, and a user evaluation assessing the prototype's usefulness, approachability, and intrusiveness.

Dedication and Acknowledgements

I would like to thank my supervisor, Dr Anne Roudaut, for her guidance, patience, and constructive feedback throughout this project. Her direction was important to shaping both the research questions and the form of the final system.

Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others including AI methods, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Luke Kim, Tuesday 5th May, 2026

AI Declaration

I declare that any and all AI usage within the project has been recorded and noted within Appendix A or within the main body of the text itself. This includes (but is not limited to) usage of text generation methods incl. LLMs, text summarisation methods, or image generation methods.

I understand that failing to divulge use of AI within my work counts as contract cheating and can result in a zero mark for the dissertation or even requiring me to withdraw from the University.

Luke Kim, Tuesday 5th May, 2026

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Ethics Statement

A compulsory section

- “This project fits within the scope of ethics application 6683, as reviewed by my supervisor, Anne Roudaut”

Supporting Technologies

The following third-party hardware and software resources were used in this project:

- **Freenove Robot Arm Kit** — the four-axis robotic arm platform on which the system is built. The kit’s Freenove control board (with onboard A4988 stepper motor drivers) is driven directly by the Raspberry Pi via GPIO.
- **Raspberry Pi 5 (2 GB)** — used as the on-robot controller. Hosts the TCP socket server that receives commands from the laptop. Drives the arm’s three stepper motors via GPIO through the Freenove board’s A4988 drivers, and the lamp-head gimbal servos via hardware PWM.
- **Open-source 3D-printed pan-and-tilt gimbal** — the two-servo gimbal mounted between the arm’s wrist and the custom lamp head was 3D-printed from an existing open-source design (<https://grabcad.com/library/gimbal-2-axis-with-sg90-mini-servo-3d-printing-1>) and used without modification.
- **Custom lamp head** — the lamp head carried by the gimbal was designed by the author in Autodesk Fusion and 3D-printed specifically for this project.
- **MediaPipe** (Google) — used for real-time human pose estimation. The PoseLandmarker Tasks API with the full BlazePose model was used to extract 33 body landmarks from each webcam frame.
- **YOLOv11n** (Ultralytics) — used for object-level detection of phones in the webcam frame. Only the pre-trained “cell phone” class is used; no fine-tuning was performed.
- **OpenCV** — used for webcam capture, frame manipulation, and overlay of debug visualisations during development.
- **NumPy** — used for vector operations on landmark coordinates during pose feature computation.
- **Python standard library** — the `threading` and `socket` modules were used to manage non-blocking robot control and TCP communication respectively.

Notation and Acronyms

ADHD	Attention Deficit Hyperactivity Disorder
HRI	Human–Robot Interaction
SAR	Socially Assistive Robot / Robotics
HCI	Human–Computer Interaction
CNN	Convolutional Neural Network
YOLO	You Only Look Once (object detection model family)
API	Application Programming Interface
TCP	Transmission Control Protocol
GPIO	General Purpose Input/Output
PWM	Pulse Width Modulation
CAD	Computer-Aided Design
PLA	Polylactic Acid
RGB	Red, Green, Blue (colour image format)
SD	Standard Deviation
DR1–DR5	Design Requirements 1–5 (Section 3.4)
RQ1–RQ3	Research Questions 1–3 (Section 1.2)

Chapter 1

Introduction

1.1 Motivation

Maintaining sustained attention during desk-based work is a challenge that affects a broad population, but is particularly acute for individuals with Attention Deficit Hyperactivity Disorder (ADHD). ADHD may experience difficulties with attention more than others with ADHD being a disorder that presents difficulties in tasks related to attention, impulsivity, and executive function Barkley (1997). ADHD is present in around 2 to 5 percent of the adult population in general Fayyad et al. (2007); Polanczyk et al. (2007). Beyond those who have formally been diagnosed with ADHD, many others also experience difficulties with attention Faraone et al. (2006); Kessler et al. (2005). Furthermore, the shift towards remote and home based work presents challenges for those who may otherwise struggle with attention. Without the ambient structure provided by shared office or classroom environments, where the presence of others naturally supports one’s ability to remain focused on their tasks, individuals must rely upon their own abilities to remain focused while working alone Wang et al. (2021). For those with ADHD or attention difficulties outside of the disorder, this lack of structure in their work environments can contribute to task avoidance, incomplete work, and increased feelings of frustration at their inability to focus Galanti et al. (2021).

For these reasons, it is helpful to introduce individuals to various productivity tools that may help individuals with ADHD to focus upon their tasks and to manage their time more efficiently. However, many of these tools share a common limitation in that they are reactive and rely upon the user’s own willingness to engage with the prompts that they provide, which can easily be dismissed or ignored. More importantly, many of these tools do not provide a social presence to the individual who is using them. One solution that has been proven to be effective in encouraging individuals to focus upon their tasks and to initiate them is the practice of body doubling, in which one works alongside another individual Eagle et al. (2024). Body doubling has been proven effective not because of any active coaching or direct instruction provided by the other person, but rather because of the ambient accountability and the sense of co-presence that another entity provides Eagle et al. (2023, 2024). However, in its traditional form, body doubling relies upon the availability of another individual who is willing to be doubled with, and this cannot be guaranteed in home or solo work contexts. As such, virtual body doubling services have emerged as a potential solution, though they introduce additional demands such as the requirement of scheduling with another individual, social obligation, and the feelings of discomfort with being observed by another individual that has prevented some from attempting to utilize these services Eagle et al. (2024).

In this project, I aim to assess whether the incorporation of a robotic companion into the body doubling practice can fulfill that same role. More specifically, as you can see in Figure 1.1, I have created an anthropomorphic robotic arm that may be placed upon a user’s desk and that will automatically turn towards the user if they become insufficiently attentive towards their tasks. This orientation behaviour will create a subtle sense of being observed by the robotic arm. The robotic arm will utilize the common design of the Pixar lamp, which is an object that is recognized by most individuals, and which possesses anthropomorphic qualities by its very nature. In general, robotic systems have been explored as companions for humans in various contexts Feil-Seifer and Matarić (2011). However, much of the current research on robots as companions has focused on physical task support, structured reminders, or therapeutic interaction. For instance, O’Connell et al. (2024) designed a robot companion that moni-

tored attention and provided non-verbal feedback to individuals with ADHD while they were performing tasks. These individuals responded positively to the intervention, though only in short sessions with the robot. Therefore, designing a robotic system that utilizes real-time autonomous detection of attentional lapses with embodied physical response and that incorporates elements of body doubling for a sense of co-presence within the user’s home environment is an underexplored topic.

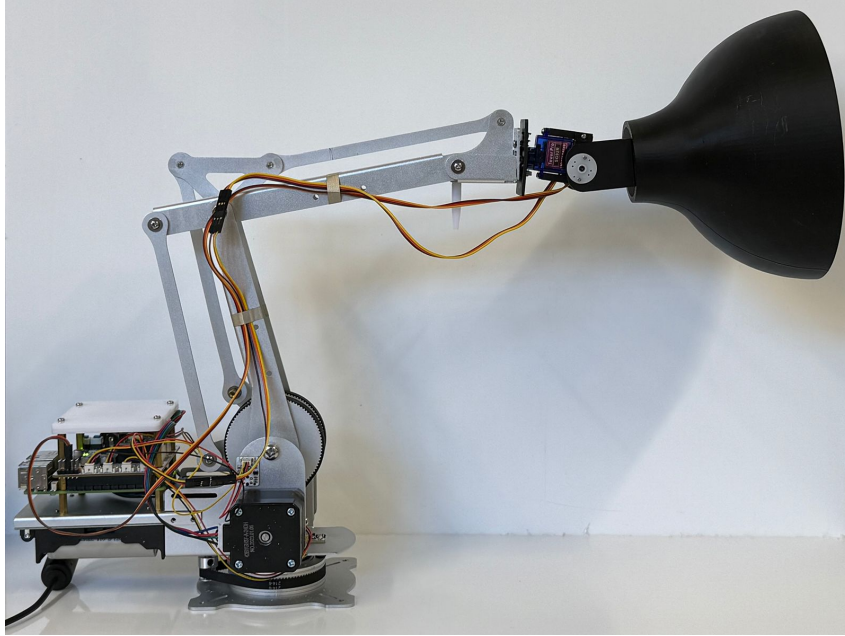


Figure 1.1: The final lamp-inspired robotic prototype in its HOME pose.

1.2 Research Questions

This project addresses the following research questions:

- RQ1. Can a vision-based pipeline (combining pose estimation and phone detection) provide a usable real-time proxy for distraction in a single-user desk environment?
- RQ2. Does the robot’s orientation behaviour constitute a noticeable and acceptable attention cue compared to a passive baseline?
- RQ3. Does a robot’s orientation-based “looking at the user” behaviour create a meaningful sense of co-presence that echoes the attention-support effect of body doubling?

1.3 Project Aim and Contributions

This project aims to design and implement a robot that can detect instances of attentional drift and respond appropriately to them with nonverbal, anthropomorphic behaviours to gently bring the user’s attention back to their task. It acts as a body doubling robotic companion.

The proposed system uses a laptop camera and a multi-cue vision pipeline that combines MediaPipe-based pose estimation with YOLO-based phone detection to identify whether a user has become distracted.

A lamp-inspired robotic arm will turn towards the user as a means of providing embodied attention without using sound, text, or interruptive features. The embodiment of the robot is character-like in form rather than human-like. Drawing on HRI research into the uncanny valley Mori (2012) and the attentional effects of robot gaze Admoni and Scassellati (2017), the system is designed to occupy a zone of moderate anthropomorphism that is both expressive enough to indicate its status as a social agent, but non-threatening enough to be comfortable with its presence in the background of their work sessions.

In summary, the project makes the following contributions:

- C1. A prototype of an autonomous attention-support robot that combines real-time distraction detection with physically embodied response, designed specifically for single-user home desk environments. As my knowledge, no prior system has integrated autonomous vision-based distraction detection with embodied orientation response in this specific deployment context. The final prototype is shown in Figure 1.1.
- C2. A distraction detection pipeline integrating pose-based monitoring (BlazePose via MediaPipe Bazarevsky et al. (2020)) and object-level phone detection (using YOLOv11n via Ultralytics), enabling the system to identify both postural drift and active device use without additional hardware.
- C3. A set of empirically derived design requirements drawn from a formative survey with twelve participants, seven of whom self-identified as having ADHD, providing user-grounded input on preferences and aversions for robotic attention support.
- C4. A user evaluation of the prototype assessing perceived usefulness, approachability, and intrusiveness, providing preliminary evidence on whether orientation-based robotic behaviour can function as an attention cue in practice.

1.4 Dissertation Structure

Chapter 2 reviews the literature across four areas: socially assistive robots for attention support, body doubling and co-presence research, anthropomorphism in HRI, and vision-based distraction detection. Chapter 3 presents the results of the formative survey and the design requirements that are obtained as a result of participant response. Chapter 4 describes the system design and implementation of the system, including its embodiment as a robot, hardware setup, vision pipeline, and robot control architecture. Chapter 5 presents the results of the user evaluation, including the methodology, procedure, and results. Chapter 6 presents the discussion of the results in relation to the research questions, as well as conclusions regarding the research, limitations of the research, and suggestions for future research.

Chapter 2

Background

2.1 Socially Assistive Robots for Attention and Task Support

Socially assistive robotics (SAR) is a subfield of human-robot interaction that is concerned with robots that provide support to humans not through physical means, but through social means. SAR influences the behaviour, motivation or cognitive state of that user through their interaction with the system.

While much of the existing literature on SARs has focused upon populations such as the elderly or children with autism spectrum disorder, some more recent attempts have begun to explore the context of everyday productivity. One of the most relevant studies to be performed on the topic was that of O’Connell et al. (2024), who created and evaluated an in-dormitory SAR study companion for students with ADHD.

In a three-week study of eleven participating students, the robot successfully monitored the students’ attention during the study sessions, providing them with non-verbal feedback. The students all tended to have positive reactions to the presence of the SAR, with most stating their desire to utilize the SAR after the study period. These findings help the user to indicate that the non-verbal and movements of the SAR were well-suited to the system, because the students were receptive to the movements of the robot and did not feel as if the robot was invasive to their efforts.

This finding is related to the present project itself, which investigates the aspects of non-verbal, orientation based interaction with a robotic companion in the desktop-based task project. The study by O’Connell et al. (2024), however, was performed only in brief and controlled settings. Thus, questions exist as to how such a system would be effective in different and more naturalistic environments over longer periods of time. The present project intends to target the environment of the individual’s home, or in their personal space, and to detect for them autonomously while they are unsupervised. Furthermore, one of the major reasons for why user might be likely to engage with a automated companion or human can support attention is a question that has arisen in the literature on body doubling.

2.2 Body Doubling and Attention Support

Body doubling refers to the idea of working alongside another person — they do not have to be working on the same task but the presence of another individual can help with the feeling of motivation to begin a task or to avoid it entirely. Though body doubling is a practice often discussed in the context of individuals with ADHD or self-help contexts regarding focus and motivation to perform a task, Despite the prevalence of body doubling in online content, it remained largely unexplored in academic literature until recently. Eagle et al. (2023) provided the first academic treatment of the phenomenon in a poster study that surveyed 220 neurodivergent participants about their experiences with body doubling. Through their study, the researchers proposed that body doubling exists on a continuum of physical and remote presence, indicating that the “effect” of body doubling can be achieved whether the individual is physically beside the student or whether they are simultaneously watching the student via media channels like “Study With Me” videos. This level of framing is significant for the present project in that a robot could potentially exist within this continuum, even without the need for social interactions from the human. The full elaboration of this work was subsequently published as a journal article: Eagle et al. (2024). In this paper, the authors report that the two most common reasons for body doubling was the performance of

tasks and the need to hold accountability with another individual, with companionship and the reduction of loneliness also being cited as common reasons for the practice. Critically, the participants discussed the benefit of body doubling in the context of simply being alongside the other individual without having to perform any tasks themselves. Thus, this aspect of body doubling indicates that a robot may not be required to interact with or communicate with the individual to provide such an effect. Presence and orientation may be sufficient. Despite its growing uptake, body doubling has not yet been incorporated into robotic systems. The present project is an attempt to operationalise the co-presence dimension of body doubling through a robot that

monitors attention and responds with orientation behaviour, requiring no human partner to be available. The form of the robot that provides this function, however, is important to consider — a discussion of anthropomorphism of the robot is included in the following section.

2.3 Anthropomorphism in Human-Robot Interaction

How a robot looks and moves has a substantial influence upon whether an audience finds it approachable, trustworthy, and worth engaging with. The concept of anthropomorphism in relation to robotics refers to the degree to which a robot is perceived as having human-like qualities. Generally, robots with some degree of anthropomorphism are found to be accepted and engaged with by end users more than robots without anthropomorphic features. This relationship was first formally described by Mori (2012). Mori proposed what would become known as the uncanny valley hypothesis, which stated that as robots become more human-like in their appearance, the humans associated with those robots tend to experience more positive emotions towards those robot creations. However, as the human likeness of the robots increases beyond some threshold, humans begin to feel uncomfortable towards those robots. While several debates surrounding this hypothesis have occurred since its publication, the general idea of the uncanny valley remains influential in the creation of robots today. Zlotowski et al. (2015) built on this foundation by arguing that anthropomorphism is a multi-dimensional construct, such that the effects of anthropomorphism depend upon the specific features that are anthropomorphized. The authors found that anthropomorphism influenced the likeability of robots, but also had an impact upon the expectations that were placed upon the robot of its capabilities and social accountability for those tasks. Furthermore, the higher the degree of anthropomorphism applied to a robot, the higher the expectations of the robot of its capabilities — many of which were potentially difficult to fulfill. Thus, the authors recommended applying anthropomorphism to assistive robots to create a moderate level of anthropomorphism rather than fully humanoid robots. At the level of specific behaviours, Admoni and Scassellati (2017) provide a review of the influence that eye gaze directed at the human in interaction with the robot has on perceptions of the robot. Such directed gaze was found to increase the likeability, naturalness, and attentiveness of the robots. Furthermore, listener directed gaze from the robot during task performance was found to improve the human participants' recall during task performance. This finding supports one of the main design decisions for the robot: to use gaze-like orientation as the primary way in which the robot is to respond to when the user becomes distracted.

When the robot turns towards the user, its movement is meant to be understood by the human. The movement is a means of communicating to the user that the robot has recognised their distraction, and is attempting to re-direct the user's attention to the task.

This finding also helped to inform the decision to utilise a lamp-inspired form for the robot. The robot's appearance allows to convey the direction in which it is "looking" without the need for a human-like face and eyes. Because humanoid robots can be somewhat uncomfortable to observe for some individuals, the design that is inspired by lamps can help to alleviate that uncomfortable observation of the robot while still retaining some of its characteristics.

However, in order for the robot to function autonomously, it must first be able to recognise when the user becomes distracted. This topic is covered in the next section of this report.

2.4 Vision-Based Attention and Distraction Detection

Being able to detect when a user is becoming distracted from the task requires some form of sensing technology. One of the most common forms of sensing technology for monitoring human attention is based upon computer vision. Computer vision algorithms enable monitoring of the user via cameras, and can determine features of the users attention through the observations of the users head pose, body posture, and movements. The foundational technique underlying the present system is

pose estimation via MediaPipe, which is built on BlazePose Bazarevsky et al. (2020). BlazePose is a lightweight CNN model that can recognise 33 different body landmarks at a rate of over 30 frames per second on a standard mobile processor, making it appropriate for use in real time in monitoring a single user at a desk-based workspace. Unlike systems that are designed to detect the poses of multiple individuals in an environment, BlazePose is optimized for situations in which only a single individual needs to be monitored, which is the context of this project alone. In addition to pose-based monitoring, the present system incorporates an object-

level phone detection module based on the YOLO architecture Redmon et al. (2016) that infers whether

the user is holding or attending to a smartphone. Phone use represents one of the most common yet disruptive forms of abandonment of tasks while performed at the desk, but not captured by body pose alone. By fusing the determination of body pose with the detection of objects in the scene, it is possible to detect when an employee is either experiencing distraction from tasks or performing another task involving the use of their smartphone — all without requiring any additional hardware to the webcam setup. Recent work has extended vision-

based sensing to attention monitoring contexts more broadly. Liu et al. (2022) developed WEDAR, a system that detects changes in the attention of e-reader users based on visual cues from their webcams, such as their head pose, gaze direction, and body movement.

More than 900,000 video frames were labelled into six different categories of visual behaviour. The study found that using several visual features allowed them to accurately detect when the attention of a user was drifting from their e-reader. Furthermore, the system utilized a general detection-feedback loop to accomplish these goals. This is the same loop that is employed by the system implemented in this project.

The authors found that head orientation and body posture were two of the best visual cues to use in detecting when a user was becoming distracted. However, most of the existing research in this area has occurred in environments like classrooms or in research studies in the laboratory. Few studies have investigated the attention of a single user who is reading on a desk in their home environment.

Furthermore, most of these existing systems provide feedback to the individuals through screens or auditory cues. However, few of these systems provide feedback through some form of physical movement. This project implements a lightweight system that can be used in a home setting with a single user while also providing feedback in the form of the movement of a nearby robot.

2.5 Research Gap

The foregoing review identifies four relevant bodies of work — socially assistive robots, body doubling, anthropomorphism in HRI, and vision-based attention detection — each of which has been developed largely in isolation. What remains underexplored is their integration into a single autonomous system designed for personal, home-based use. Existing SAR research for ADHD support has demonstrated user acceptance of non-verbal robotic feedback O’Connell et al. (2024) but has not addressed real-time, autonomous distraction detection. Body doubling research has established the value of passive co-presence Eagle et al. (2023, 2024) but has not explored robotic embodiments of this concept. Vision-based attention monitoring has produced effective detection pipelines Bazarevsky et al. (2020); Liu et al. (2022) but has not been paired with physically embodied responses. And HRI research on gaze and anthropomorphism has shown that orientation behaviour is a powerful social cue Admoni and Scassellati (2017) but has not been applied specifically in the context of attention redirection for neurodivergent users. The present project sits at the intersection of all four areas. It proposes a system that uses lightweight pose estimation to detect distraction in real time, and responds through the orientation behaviour of a character-like robotic companion — combining the social presence of body doubling, the legibility of anthropomorphic gaze cues, and the technical tractability of vision-based monitoring in a form designed for everyday home use.

Chapter 3

Preliminary study

Most productivity tools that aim to help people focus on their tasks are mainly based on screen-based reminders and alarms. While these tools are available to users, they can be either intrusive due to the need for repetition to remind the users of the tasks, or can be ignored altogether by the users. It is especially strong for those who have difficulties with attention themselves. However, a robot that is physically present with the individual may be able to provide some support in ways that are not limited to text or sound. However, before actually designing such a robot, there were some things that needed to first be confirmed. To understand how the target users of the product perceive the intervention of robots in their tasks, what types of behaviour they enjoy or find uncomfortable, and whether the concept of body doubling from robotic assistants is meaningful to those users. Thus, it was necessary to perform the formative survey. The purpose of the formative survey was to uncover patterns in the users that would help to inform the product design of the robots.

3.1 Participants

The survey was distributed online via Google Forms and completed by 12 participants. This sample size is consistent with established guidance on qualitative research. Since the purpose of the survey itself was to identify design directions rather than produce statistically generalisable findings, it also aligns with the scope of formative HCI studies (Guest et al., 2006; Vasileiou et al., 2018). Of the 12 participants, seven were self-diagnosed with ADHD and did not have a formal diagnosis. Five did not identify as having ADHD but responded that they had attention or task-initiation difficulties. Age range was from 18 to 54 years of age with the majority of respondents falling between the ages of 25 and 44. Eight of the participants selected female as their gender and four selected male as their gender.

3.2 Survey Design

The questionnaire included 41 items combining Likert scale and multiple choice questions as well as open ended questions in order to provide thorough responses to the participants. The questionnaire was conducted in conjunction with two other students who are also researching ADHD. The questions that I created, items 32–41, included four different themes regarding the participants' experiences with body doubling, openness to a co-working robot, preferences and aversions regarding specific robot behaviours during distraction events, and their opinions of robotic partners based upon their appearance.

3.3 Results

3.3.1 Body doubling: prevalence and mechanism

Of the 12 participants that completed the survey, nine (75%) reported that they had attempted to body double at least once, and that their experience with body doubling was effective. Of those who reported having attempted to body double, the most common reasons provided for doing so were ease of initiating tasks (7/9) and increased feelings of accountability (7/9), followed by reduced feelings of distraction (4/9) and loneliness or overwhelm (4/9). as shown in Figure 3.1 These findings are in agreement with the survey of 220 neurodivergent participants that was reported by Eagle et al. (2024). The authors found that the

two main motivations for attempting to body double another participant was the ease with which tasks can be initiated while body doubling, as well as the feelings of accountability that participants have while body doubling. Furthermore, the researchers determined that the benefit of body doubling is not based upon any particular communication behavior by the body doubled participant, but instead in the passive social presence of that other entity. Thus, these findings suggest one potential benefit of employing a robot in place of a human for body doubling efforts.

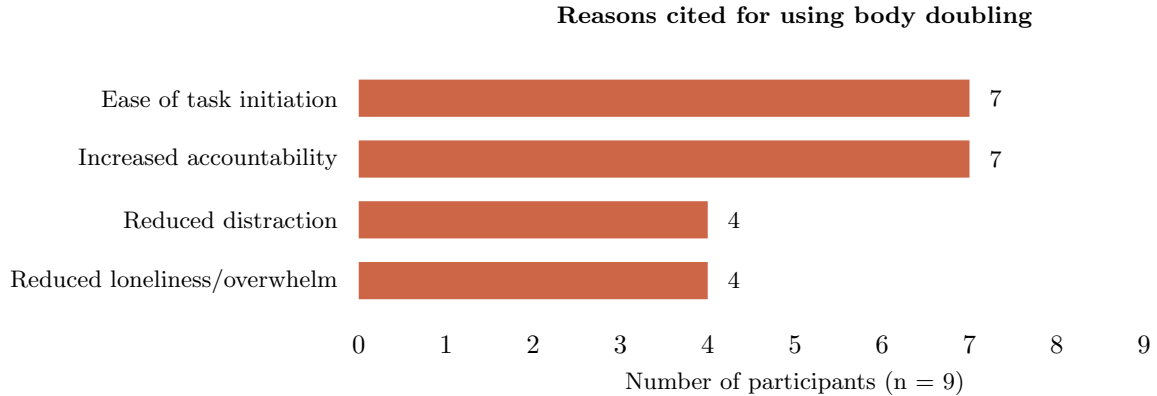


Figure 3.1: Reasons cited by participants for engaging in body doubling (n = 9 who reported prior experience). Ease of task initiation and increased accountability were jointly the most frequently cited motivations, each endorsed by 7 of 9 participants.

3.3.2 Openness to a co-working robot

When I asked the participants about the likelihood of them finding a robot that would co-work alongside them to be helpful, on average they rated such a robot as having a helpfulness of 3.25 out of five (SD = 1.06**; see Figure 3.4**). Furthermore, some of the qualitative responses support these findings. For example, one participant commented: “I am quite curious about a robot clearly looking at me. I think it may have some effect.” Another participant mentioned that they would like to see a robot with the appearance of the Pixar film’s lamp. Thus, while not all participants were eager to have a robot in their homes, they did find the concept to be credible.

3.3.3 Preferred and rejected robot behaviours

Participants were asked to rate the four responses to distraction. Responses included subtle movement, annoyed movement, looking at the user more obviously, and online streaming to a body doubling website. The results of the survey showed a difference in the type of behaviour of the robot that was liked by the participants as compared to the behaviour that was disliked by the participants. The behaviour that was liked by the majority of participants was the behaviour of the robot to “look at the user more obviously.” In this case, nine of the twelve participants indicated that this behaviour was either “helpful” or “very helpful.”

In contrast, the behaviour of the robot to perform an “annoyed movement” was the least accepted response by the participants. In this scenario, seven of the twelve participants indicated that this response to the participants’ distraction was “not helpful.”

Figure 3.2 summarises the full distribution of ratings. These results, therefore, indicate that it is best for the robot to respond to the distraction of the participants in a calm and attentive manner, as opposed to an emotional manner. Thus, the design of the system to respond to the participants’ distraction with a turn of the robot towards the participants indicates a rejection of providing such emotional or annoyed behaviours.

3.3.4 Aversive behaviour patterns

Next, the participants were asked which behaviours would make them feel uncomfortable or not be helpful with the robot. The most selected answer was that loud noises and sudden movements would be uncomfortable for the participants. Eleven of the twelve participants selected this answer. Other

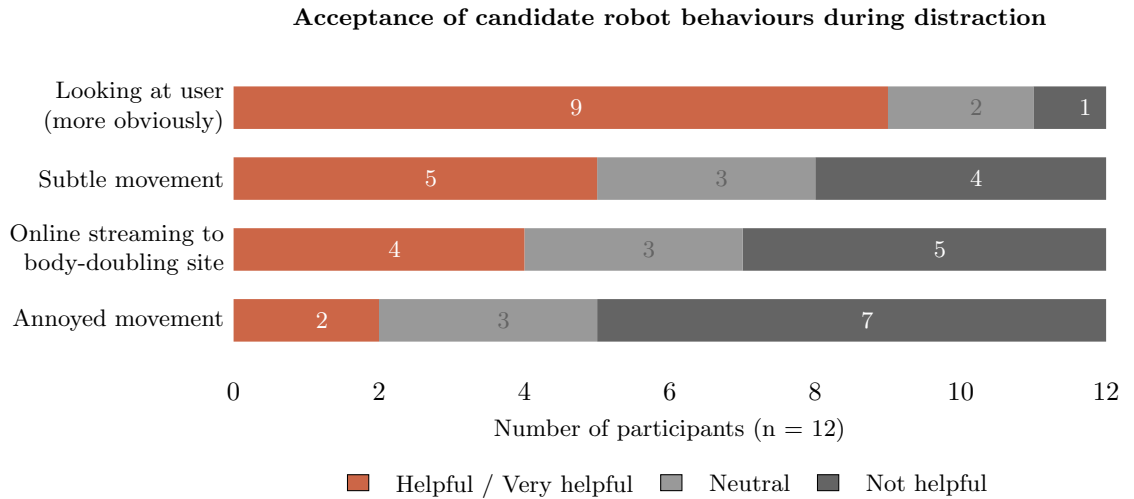


Figure 3.2: Participant ratings of four candidate robot responses to distraction (n = 12). “Looking at user (more obviously)” was rated helpful or very helpful by 9 of 12 participants, making it the most positively received behaviour. “Annoyed movement” was the least accepted, rated not helpful by 7 of 12.

behaviours that were strongly rejected by the participants included those that would cause too much talking or interruption by the robot(chosen by ten of the twelve participants), as well as those behaviours that would result in excessive notifications from the robot (chosen by nine of the twelve participants). Thus, participants indicated a desire for the robot to avoid being too noisy or demanding of their attention.

Five of the twelve participants also indicated discomfort with behaviours that used a critical or pressuring tone for the robot, as well as those that exhibited overly expressive movements. The full pattern of rejected behaviours is shown in Figure 3.3. Thus, these behaviours were less desired by the participants than the others.

Overall, then, the participants indicated a desire for the robot to avoid any behaviours that are sudden, loud, noisy, interruptive, demanding, or emotionally intense. Instead, the robot is to exhibit a calm and subtle behaviour when performing its functions.

Three participants noted that feeling watched or judged would be uncomfortable. It suggests that while attentive robot behaviour get positive feedback, it must be calibrated to feel supportive rather than feeling being watched.

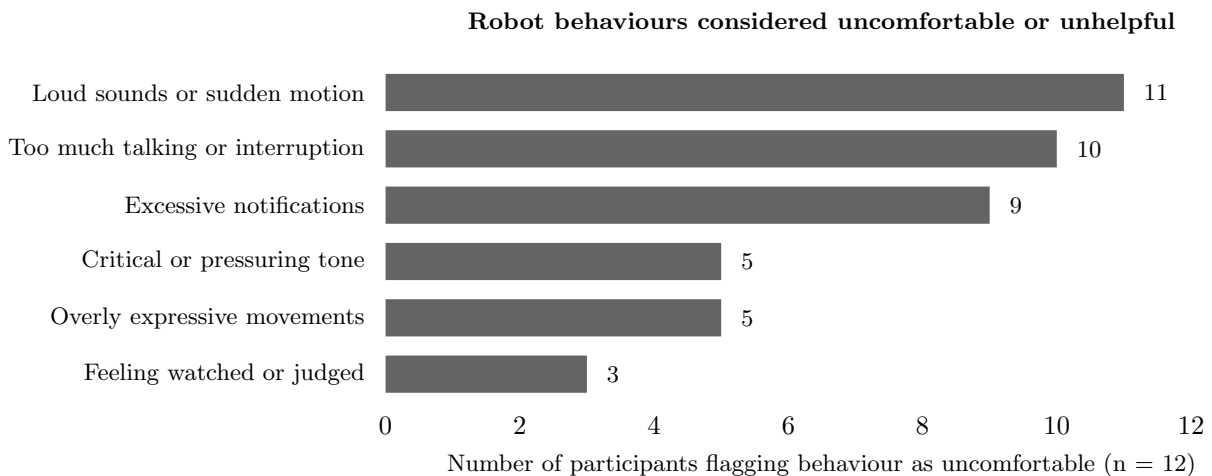


Figure 3.3: Robot behaviours considered uncomfortable or unhelpful by survey participants (n = 12). Loud sounds or sudden motion was the most widely rejected, flagged by 11 of 12 participants, followed by too much talking or interruption (10/12) and excessive notifications (9/12).

3.3.5 Appearance and acceptance

Participants were also asked whether the robot’s appearance would have an effect upon their willingness to use the robot. Specifically, the question was whether the robot would be more likely to be accepted if it had a cute and familiar appearance. The average score was 3.9 out of 5, with six participants providing the highest possible score (Figure 3.4).

The fact that appearance was discussed in this study indicates that it was an important factor for this particular group of users because of it gets high ratings. The participants were not just considering whether the robot performed its intended tasks, but also whether it would be comfortable with the presence of the robot.

For these reasons, the choice of a character-type design for the robot was supported by the results of this study.

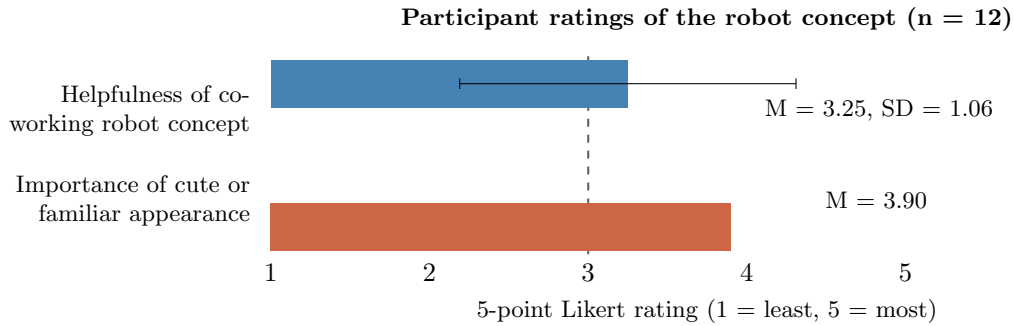


Figure 3.4: Mean Likert ratings for two key survey items (n = 12). Importance of a cute or familiar appearance was rated highest (M = 3.90), while openness to a co-working robot concept showed moderate acceptance (M = 3.25, SD = 1.06). The dashed line indicates the neutral midpoint (3).

3.4 Derived Design Requirements

The findings of the survey were used to formulate five requirements regarding the design of the robot. Each of these requirements will inform the design and implementation of the robot.

DR1 — Use gaze-based orientation as the main attention cue. The response that was most positively rated was the concept of the robot “looking at the user.” Nine of the twelve participants indicated that this behavior would be considered as either “helpful” or “very helpful.” As such, the robot’s main attention cue should be a physical orientation movement toward the user, communicating attentiveness through gaze-like movement rather than sound or text.

DR2 — Avoid any behavior that is aversive or pressurizing towards the users. Many of the negative responses indicate that behaviors such as appearing “annoyed” were considered not to be helpful by 7 of the 12 participants. Loud sounds were flagged by 11 of the 12 participants, excessive talking by 10 out of 12 participants, and using a critical tone by 5 out of 12 participants. Thus, any behavior that may feel pressurizing or negative towards the participants should be avoided by the robot.

DR3 — Use a character-like and approachable form. The high rating for appearance importance (M = 3.90 out of 5, with 6 of 12 participants giving the maximum score) supports the use of a non-humanoid but expressive form. For this reason, lamp-like robot design was selected. This form provides a clear gaze direction and a familiar looking without relying on an overly humanoid appearance.

DR4 — The robot should only be able to “communicate” through its movements. The rejection of behaviors like talking to the users, interrupting, and using notifications indicates that the robot should not use any means of communicating other than its movements, keeping the interaction ambient and non-disruptive.

DR5 — The robot should feel supportive towards the users, not surveillant. While most participants indicated positive responses towards having a robot that is attentive to the users, some

of those same participants also indicated that they would feel uncomfortable when they feeling watched or judged. Thus, after the users have returned to work, for instance, the robot should be designed to feel like co-presence rather than monitoring

Chapter 4

Implementation

4.1 Design Philosophy

The system is designed as a tool that both senses the user’s environment and takes action in response to that environment. Such design considerations are reflected in three design principles for the system.

The intervention should be physical, not informational. While many existing forms of productivity tools provide various forms of information (text, sound, visual notifications), physical movement is the only form of communication that is provided by the system. This decision was made based upon both DR4 (that any form of audible or visual communication is poorly received) and the survey findings that reflected this.

The system should be autonomous. The system should require no input from the user in order to provide an intervention. This distinction of the system from existing services that provide virtual body-doubling is one of the main reasons to utilise vision-based sensing to recognise the user’s state instead of self-reporting.

Finally, the intervention is expected to be transient and supportive rather than persistent and surveillant. The system should return to its neutral pose as soon as the user returns to a focused state; this minimises any feelings of being monitored by the system. This relates to both DR5 (that some individuals feel uncomfortable with being watched) and the findings of the survey that reflected these issues.

4.2 Translating Design Requirements into System Decisions

The five design requirements that were determined through the formative survey are mapped to system decisions in Table 4.1.

DR	Source finding	Resulting design decision
DR1	Gaze-based orientation as primary cue. “Looking at the user” was selected as the most helpful cue by 9 of the 12 participants.	The intervention’s only movement is from a neutral pose to a pose that indicates it is “looking at” the user.
DR2	Exclusion of aversive behaviour. Annoyed movement was rejected by 7/12 participants; loud sounds were rejected by 11/12.	The robot never uses buzzers, LEDs, or sudden movements; its movements are always slow and predictable.
DR3	Character-like, approachable form. A “cute” or familiar appearance was rated 3.9/5 in importance for a virtual teammate.	The design of the robot is inspired by Anglepoise lamps.
DR4	Movement-only communication. Talking and notifications were rejected by 10/12 and 9/12 participants respectively.	No audio output from the robot; no visual or screen-based notifications.
DR5	Supportive, not surveillant. 3/12 participants were uncomfortable with feeling watched.	The system returns to its neutral pose automatically once the user has returned to a focused posture.

Table 4.1: Mapping of design requirements to system design decisions.

Each of these design decisions informed the remaining elements of the system, which are described in the sections that follow.

4.3 Robot Embodiment

The embodiment of the robot was determined to have three design considerations that had to be satisfied: its need to be legible, approachable, and to avoid the uncanny valley.

A humanoid robot that possessed a face was rejected as a potential embodiment of the system due to each of these three considerations. A face and a set of eyes are features that indicate to users that the system has the capabilities of a biological being; it is expected to be able to express emotions, to have a “gaze” and “attention span,” and it may mimic the biological being that is to be observed. Furthermore, having any resemblance to a biological being is correlated with feelings of being watched and surveilled — the exact feelings to be avoided by the system. Finally, humanoid faces are recognised by many as potentially uncanny; while the uncanny valley is a difficult concept to define, exposure to humanoid faces (especially when they are different from those of the observer) can induce feelings of discomfort in users Mori (2012).

An abstract form for the robot, such as a screen with an animated dot, was also rejected. While such a screen-based embodiment may seem to satisfy the requirement that the robot should be “approachable,” one of the main findings of the body-doubling literature is that users require the physical presence of the observed entity in order to experience feelings of support and reduced fatigue Eagle et al. (2024).

The Anglepoise lamp embodiment of the robot was the chosen form for the system. This embodiment exists in the middle ground between these two extremes. As a familiar object in a home, the robot is immediately legible and approachable. Furthermore, the design of the lamp allows the system to “look at” the user, which is its primary intervention; it does not require a face or a set of eyes. Additionally, a participant in survey referenced a Pixar lamp inspired design as an appealing direction. It indicates that this embodiment will be directly accessible to its target users.

The Freenove robot arm kit was chosen as the main platform upon which to implement the robot. The kit’s mechanical layout has a base, two jointed arms, and a wrist-mounted end effector. It approximates the Anglepoise-inspired embodiment of the system, but the kit does not include a any lamp-shaped head. The lamp head was therefore modelled in Autodesk Fusion and 3D-printed for this project. To give the head second axis of expression beyond the arm’s overall orientation, a two-servo pan-and-tilt gimbal was placed between the arm’s wrist and the lamp head. the gimbal was 3D-printed from an open-source design and used without modification. Fabrication of the lamp head and the selection of final print parameters are described in Section 4.7.

4.4 System Architecture

The architecture of the system is shown in Figure 4.1.

The main components of the system are arranged in such a way as to meet two main design considerations.

First, the laptop is used for its high computational power, relative to the Raspberry Pi. The process of detecting the user’s posture, recognising when they are engaged in a distracting activity, and outputting a movement command is computationally intensive; performing these processes on the Raspberry Pi would introduce latency that would prevent the system from effectively providing the required intervention.

Second, the physical elements of the system can be located separately from the laptop. This consideration is important in the context of home environments. Users may wish to place the robot in various locations within their homes, allowing them to have the assistance of the robot wherever they are performing their focused tasks.

4.5 Interaction Model

The robot’s intervention is built around two arm poses which are home and look. And two short head motions produced by the gimbal, shake no and nod yes, illustrated in Figure 4.2.

In home, the lamp head points away from the user. But in look, the head is oriented toward the user. shake no is a brief side-to-side head motion, and nod yes is a brief vertical head pitch.

A complete intervention cycle returns as follows. When sustained distraction is detected, the arm moves from home to look, and upon arrival at look the gimbal performs a brief shake no to warn the moment the robot has registered the user’s distraction.

When the user returns to a focused posture, the arm moves back from look to home, and upon arrival the gimbal performs a brief nod yes to get users compliment the user’s return to task.

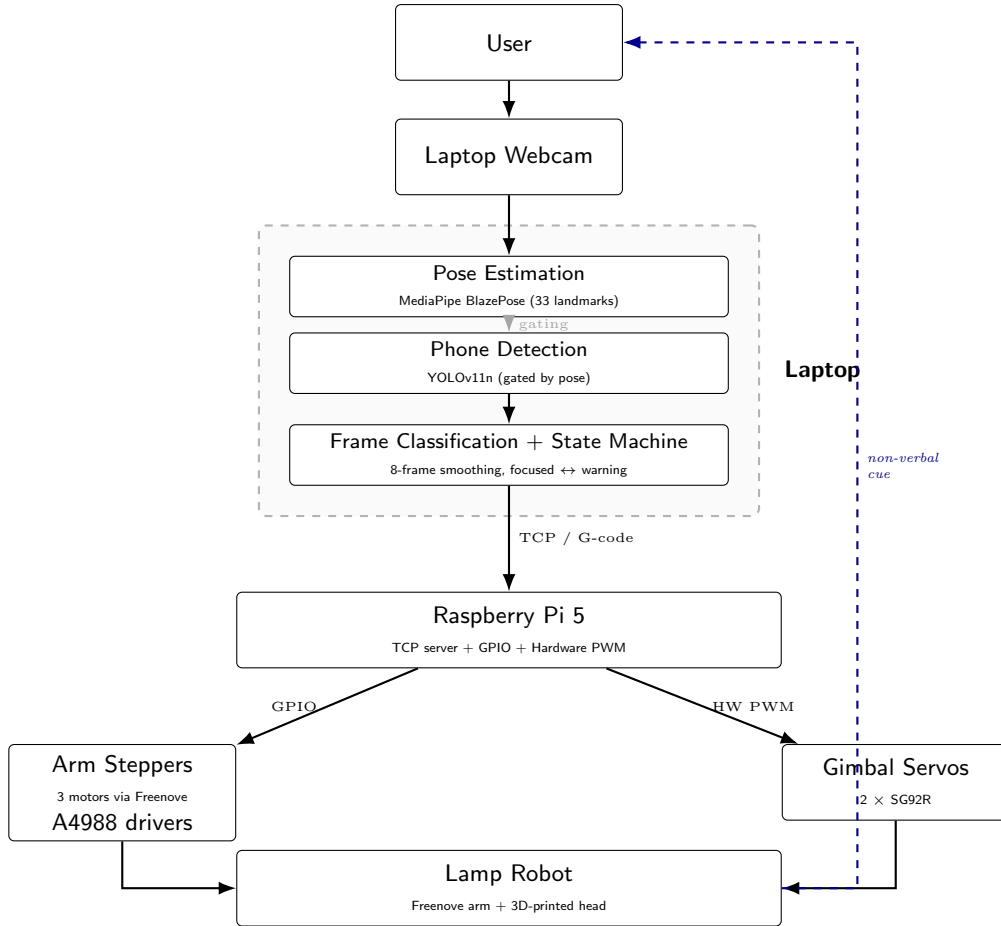


Figure 4.1: System architecture. The laptop captures webcam input and runs the vision pipeline—MediaPipe BlazePose pose estimation gated to YOLOv11n phone detection—together with the distraction state machine. Movement commands are dispatched as G-code over TCP to a Raspberry Pi 5, which drives both the arm’s stepper motors (via the Freenove control board’s onboard A4988 drivers) and the gimbal servos (via hardware PWM) directly through its GPIO pins. The physical orientation of the lamp robot then provides a non-verbal cue to the user, closing the feedback loop.

The intervention should be an orientation-based behaviour enhanced with brief head motions at the transition boundaries, rather than a static gaze. There are no other interventions within this category such as with no escalation, expressive variation, or context-dependent gestures. This decision is based upon the findings of the survey of the participants. Users indicated the most positive reaction to a consistent form of intervention (DR5), and any form of intervention that could be read as escalation was rejected (DR2). The gimbal motions are kept short and subtle, and are gated by a 10-second cooldown to prevent rapid repetition, in line with DR2’s exclusion of aversive behaviour and DR4’s restriction to movement-only communication. Restricting the intervention to a single fixed cycle also supports predictability and trust over extended use, in line with broader guidance in socially assistive robotics that interaction models for everyday assistive use should prioritise predictability and minimalism over expressiveness Feil-Seifer and Mataric (2011).

4.6 Perception Strategy

The primary strategy for recognising distraction of the user from their tasks is based upon three main considerations:

- the system must utilise only the laptop’s webcam to observe the user;
- the system must work at interactive speeds on consumer-grade computer hardware;

- the system must recognise forms of distraction that are easily distinguishable so that the intervention can respond appropriately.

The perception strategy for the system is based upon pose-based recognition. The reasons for this strategy are that poses can be recognised at the distances from the laptop to the user that are typical for desk-based work; body pose estimation libraries are readily available and include detection of wrist positions (which can be used to recognise if the user is using their phone); and that the entire strategy is autonomous, requiring no input from the user to recognise their state.

More specifically, the pose estimation library used is MediaPipe’s PoseLandmarker Tasks API with the model that returns the 33 points of the estimated pose; these include the wrist positions needed for the detection of users that are potentially using their phones Bazarevsky et al. (2020). The rationale for using this library is that it is optimised for the detection of a single person, can work in real time on CPUs, and includes pose features that can determine if the pose of the user indicates phone use.

Pose-based recognition alone is not a reliable means of distinguishing between cases in which the user may be looking at their phone and those in which they may be looking at their notes or some printed document. Therefore, a separate stage for the detection of phones is initiated in cases in which the pose model indicates that the user may be looking at their phone. This two-stage model ensures that both the requirement of performance at interactive speeds (because the object detector is only called during the recognition of phone use) and accuracy in recognising distraction (due to the use of additional means of identifying the presence of a phone) are satisfied.

4.7 Hardware Setup

The system is comprised of four main hardware components. The main laptop performs the computation processes that determine if the user is distracted or not. it captures its own webcam feed, performs all of the pose recognition and phone detection, and runs all of the logic that determines how the system have to respond.

Additionally, the laptop also acts as the SSH client that allows it to communicate with the Raspberry Pi. it sends the commands that tell the Raspberry Pi when the robot’s arm should move.

The four components are shown in Figure 4.3. The Raspberry Pi 5 (2 GB) hosts a server that listens for commands from the laptop. When it receives a G-code command to move the arm, it parses the command and signals to the Freenove control board’s onboard A4988 stepper drivers via GPIO to drive the three arm stepper motors. The Pi additionally drives the gimbal servos directly via hardware PWM. It uses the same board but a separate signalling path. This unified controller setup keeps wiring simple and removes the latency that an additional microcontroller would launch.

The Freenove robot arm is the physical embodiment of the system. It is positioned in a fixed location next to the user’s main workstation, and is oriented in a way that its movements do not interfere with the user’s view of their screen. Two-axis pan-and-tilt gimbal is mounted at the arm’s wrist. it is 3D-printed from an open-source design¹ and used without geometric modification. Two SG90 micro servo motors actuate the pan and tilt axes. Each is rated at approximately 1.6 kg·cm of stall torque. The lamp head is mounted to the upper plate of the gimbal.

4.7.1 Lamp Head Fabrication

The lamp head was 3D printed in PLA using a Bambu Lab P1S 3D printer. Although the CAD file was authored once for the design, three print iterations were made before the assembly satisfied the mechanical constraints of the rest of the hardware.

Two independent torque budgets had to be respected. The shoulder and elbow joints of the Freenove arm must be able to hold the head against the force of gravity while the head is horizontally offset from the base of the arm. Second, within the gimbal itself, the two SG90 servos must actuate the head about the pan and tilt axes against the same gravitational load. Each of these constraints had to be satisfied simultaneously for the design to be considered successful.

The first iteration (v2) utilized the default Bambu Studio profile with a 0.20 mm layer height, 15% grid infill, two perimeter wall loops, five top and three bottom shell layers, and tree supports generated at a 30-degree overhang threshold. The slicer indicated a mass of 140.91 g. When the printed head was bolted to the gimbal, and the gimbal to the arm wrist, the total mass was too heavy for the arm to hold stable on its own; the shoulder and elbow joints sagged, and the arm could not hold the look pose.

¹<https://grabcad.com/library/gimbal-2-axis-with-sg90-mini-servo-3d-printing-1>

The binding constraint at this mass was the arm’s joint torque, well before the gimbal servos were even commanded to move.

For the second iteration (v3), the layer height was reduced to 0.16 mm and the infill was changed from a 15% grid to a 10% gyroid pattern. Results: Gyroid infill provides equivalent or better isotropic stiffness at lower densities than grid infill, reducing the amount of infill material needed without compromising the structural integrity of the printed component. The parameters for the walls, shells, and supports were left unadjusted. The slicer predicted a mass of 134.49 g, which is a $\sim 5\%$ reduction compared to v2. This mass is within the joint torque capabilities of the arm, and the arm is able to hold the gimbal and head in place in the look pose without sag. However, when the gimbal servos were then commanded to perform their pan and tilt motions, they stalled — the SG90 servo motors were not able to lift up the head against the influence of gravity. The binding constraint had shifted from the arm joints to the servos themselves.

The third iteration (v4) targeted the servo torque margin specifically. The infill density was reduced further to 5% gyroid, walls were reduced from two to one, the top shell count was reduced from five to three, and the bottom shell count was reduced from three to two. The layer height was held at 0.16 mm. The slicer software indicated a mass of 84.40 g, which is a reduction of 37% from v3. At this mass, the arm can hold the head steady at look while the SG90 servos are able to move each gimbal axis to any position within their physical range of motion. The v4 configuration was the version used in all subsequent integration tests and in the user evaluation presented in Chapter 5. The three iterations of the system are summarized in Table 4.2; the masses for each of these iterations are that were predicted by the slicer software, rather than the actual weights of each of the printed parts.

The v4 print settings are close to the lower bound of structural integrity for this geometry. Due to the use of only a single perimeter wall and 5% infill, the printed head is less rigid than the v2 or v3 versions and would be unlikely to endure through handling outside of a controlled study environment. A future revision would perhaps more sensibly target mass reduction through changes at the CAD level, such as reducing the outer dimensions of the head, hollowing internal regions, or adding internal ribs — rather than changing the slicer parameters to the minimum values.

Ver.	Layer	Infill	Walls	Top/Bot	Mass	Outcome
v2	0.20 mm	15% Grid	2	5/3	140.91 g	Arm joints could not hold the assembly; visible sag at LOOK.
v3	0.16 mm	10% Gyroid	2	5/3	134.49 g	Arm holds steady; gimbal servos stall under load.
v4	0.16 mm	5% Gyroid	1	3/2	84.40 g	Arm and servos both within envelope; final configuration.

Table 4.2: Summary of the three lamp head print iterations. Mass values are predicted by the Bambu Studio slicer.

4.8 Vision Pipeline

4.8.1 Pose Estimation and Calibration

Pose estimation uses the PoseLandmarker Tasks API that comes with MediaPipe’s pose estimation library. More specifically, the model that returns the 33 points of the estimated pose (including the wrist positions used to detect if the user is using their phone) is used. This model is run on every frame of video that is captured by the laptop.

When the system is initially started, a period of calibration is performed. During this calibration period, which lasts for 10 seconds, the user is prompted on the screen to assume their usual posture while performing focused work. During this calibration period, four reference values are calculated. These are the baseline Y position of the nose, the X position of the midpoint between the shoulders, the Y position of the midpoint between the shoulders, and the shoulder width. These measurements are used later in the system to normalise the measurements of features of the user’s pose; different users will have different shoulder widths, for instance, and the pose model’s measurements would otherwise be less adaptable to different users. Furthermore, these measurements allow the system to avoid requiring re-calibration each time that the user may change their position relative to the laptop.

4.8.2 Pose-Based Feature Computation

For each frame of video, three features regarding the user’s pose are computed. These features are used to help determine if the user is engaged in any form of distraction.

The looking away feature is recognised when the deviation of the user’s nose from the midpoint between the user’s shoulders is more than 0.18 units in either direction (the deviation is divided by the user’s shoulder width to permit for differences in shoulder width); it is also flagged when the user’s entire torso is more than 0.10 units in either direction from the centre of the screen. This feature ensures that the system recognises if the user is turning away from their screen (which may indicate that they are focusing upon some other task), and also captures cases where the user’s pose may be interpreted as looking at a screen on a different device.

The looking down feature is recognised when the position of the user’s nose is more than 0.06 units in the Y direction below their baseline Y position; this feature is triggered when the user is looking at their notes, another screen, or their phone.

The torso shift feature is recognised when the user’s shoulders are more than 0.08 units in either the horizontal or vertical direction from their baseline shoulder position; this feature recognises if the user is slumping over, for instance.

A filtering step is also performed on each frame for landmarks that are not visible to the webcam. MediaPipe assigns a visibility score to each landmark that is displayed with less than 50% confidence. Landmarks with less than 50% confidence are not used to evaluate the user. In addition, frames that do not contain pose features for the user’s nose, left shoulder, and right shoulder are completely removed. This is because the user may not be fully within the webcam’s field of view.

4.8.3 Phone Detection

Within each frame of video that is captured by the laptop, an object detection model is used to recognise if any object in the frame is a phone. More specifically, a YOLO object detection model (YOLOv11n) Redmon et al. (2016) is used; larger versions of this model were rejected due to the latency they introduced between pose recognition and phone detection.

In cases in which pose detection recognises that the user’s pose may indicate phone use (with the head down, with one or both wrists near the torso, or when the torso is shifted from a vertical posture), the YOLO object detection module is called. This gating strategy helps to ensure that the object detection is only invoked during those instances in which it is needed.

When the object detection model is called, it analyses each frame and returns any objects within the frame whose class is “cell phone”. Any phones that are within 50 pixels of either wrist landmark are recognised as being used by the user (to allow for the user’s phone to be slightly off from the wrist).

4.8.4 Multi-cue Feature Fusion and Frame-Level Classification

In each video frame, the detected features such as looking away, looking down, torso shift, and phone detection, are fused into a single label for that frame, describing what type of distraction is occurring.

The feature with the highest priority indicates that the user is engaged in a “possible phone” posture; the feature with second priority is whether the user is looking away from their screen; the third and fourth features are “looking down” and “torso shift” respectively.

Additionally, a smoothing filter is applied to each frame. The classification stage applies a filter that takes into account the classification results of the past eight frames; the label that appears in at least half of those eight frames is applied to the current frame. This strategy prevents any single frame result from altering the user’s classified state, and ensures the stability of that classification. A similar temporal smoothing approach is used in WEDAR Liu et al. (2022). where majority-vote filtering over a sliding window was found to suppress spurious frame-level noise in webcam-based attention classification.

4.9 Distraction Detection State Machine

The smoothed frame-level state is fed into a timing-based state machine to determine whether the user has been distracted for long enough to elicit a response from the robot. There are three thresholds that, when met, trigger the system to enter the warning stage of operation. If the user is observed in a looking away state for 2.0 seconds, the robot turns toward the user. Phone detection is always activated regardless of the pose state. If a phone is detected within the frame for 0.5 seconds, system sends warning.

Additionally, if no pose landmarks are detected for 1.5 seconds, indicating that the user has left the desk, system also triggers a warning. Each of these timers immediately resets when the condition is removed. These time thresholds were decided through development testing. Shorter thresholds resulted in false positives, while longer thresholds resulted in missed detections.

4.10 Robot Control and Communication

The decision to use Python threading to control the robot was made to avoid the “freeze” in the system’s main loop that would occur if a movement command were sent to the robot while it was already performing another movement. When the system detects that the user is distracted, a daemon thread is started that moves the robot towards the user; control then returns to the main loop immediately so that pose detection continues uninterrupted. Commands are sent from the laptop to the Raspberry Pi over a TCP socket connection using G-code strings, which the Pi parses and dispatches to the stepper drivers via GPIO. **Threading.Lock** protects robot busy flag. It ensures that no additional movement commands are dispatched while robot is moving.

Two reference poses are used: home (G0 X0.0 Y160.0 Z220.0) is the resting position with the lamp head pointing away from the user, and look (G0 X140.0 Y90.0 Z265.0) is the position in which the lamp head faces the user. These coordinates were determined through experience of robot physical testing.

Instead of issuing a single direct jump from home to look, the system interpolates through two intermediate waypoints. Its look via1 (G0 X43.0 Y143.0 Z235.0, the one-third point) and look via2 (G0 X107.0 Y116.0 Z250.0, the two-thirds point). Two points have a 0.8-second pause at each waypoint before the final 4.0-second motion to look. It provides more smooth and legible movement than a single direct motion. It makes robot design satisfies DR2’s requirement that movement be slow and predictable than sudden. A reduced servo speed setting is issued before this motion begins. In contrast, This is because the return is not intended to provide any cue to the user. A quick return for the next cue is more important. After the robot finishes moving, the system always waits 4.0 seconds before internally updating the pose state. This is because if the system sends the next command before the robot’s physical movement has actually finished, the logic may become twisted. When the user returns to a focused state, the system detects this and returns the robot to the home position. This method was designed to remove the feeling that the robot is intentionally continuing to monitor the user. The moment the user returns to work, the robot also returns home in order to give the user the feeling that it is a supportive presence. This directly implements design requirement DR5: the principle that the system should feel like a supporter rather than a surveillant observer.

In addition to the arm motion, a small two-servo gimbal mounted at the lamp head produces brief head motions. When the robot moves to the LOOK position, the gimbal performs a brief shake no to indicate to the user that they are in a distracted state. upon returning to home, it performs a small nodding motion to tell users returned to focus. Due to the 10-second cooldown, these supplementary motions cannot be triggered in rapid succession. It is kept deliberately short and subtle in line with DR2’s exclusion of aversive behaviour and DR4’s restriction to movement-only communication.

4.11 Limitations and Engineering Constraints

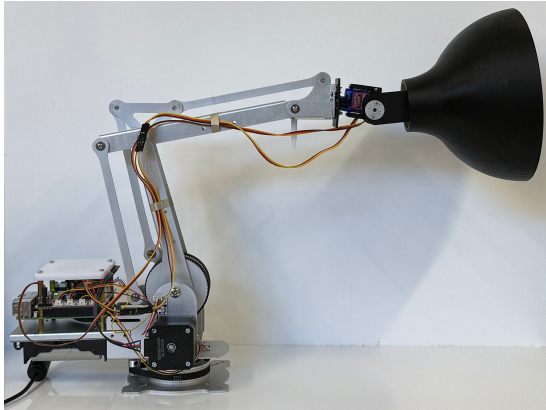
The following aspects of the system are yet to be implemented.

A new TCP socket connection between the laptop and the Raspberry Pi is established each time a command is sent to the robot. While this implementation minimises the potential for errors, it does introduce a small amount of latency in the movement of the robot, and the connection may fail if the laptop and robot are not connected to the same wireless network. Creating a persistent socket such as heartbeat signal would allow the system to avoid these kind of errors.

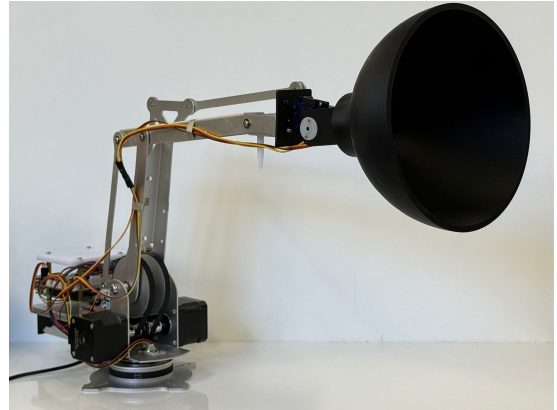
The problem with the current system is that the coordinate value of the robot’s posture to “LOOK” is hard-coded. In other words, whenever the robot’s position or the user’s position changes, the new coordinates must be measured again and put into the code. There is no problem in operating as long as the robot and the user are placed in fixed position. However, the ideal system is for the webcam to calculate the user’s location and the robot to automatically look at the user. For this to happen, the model needs to measure the offset between the predetermined robot and the laptop position. In addition, very precise tasks such as developing a model that calculates the exact coordinates of the user’s eye position using a 3D depth camera and calculating reverse electromagnetism for the robot to move to that position are required. I think these tasks are considered as descope of this project.

Using the YOLO model for phone detection can increase the processing burden on laptops. Since YOLO is always running in the current code, this may reduce the frame rate of the entire vision pipeline. One way to solve this problem is separating the YOLO detector from the main process and run it as a separate process, or transfer it to Raspberry Pi altogether to disperse the processing burden.

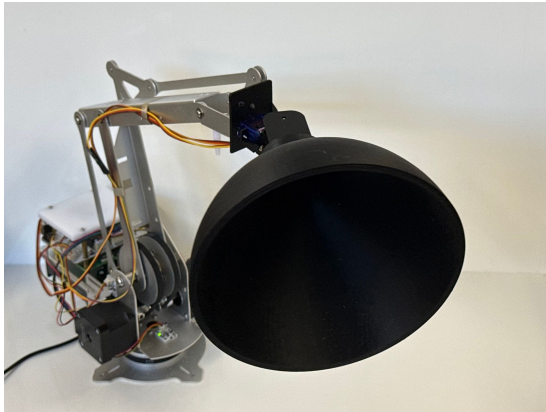
The current system only works for one user sitting in a specific location in front of the laptop. To support multiple users, the system needs to be redesigned in a more flexible way that can adapt itself to changes in the user's location.



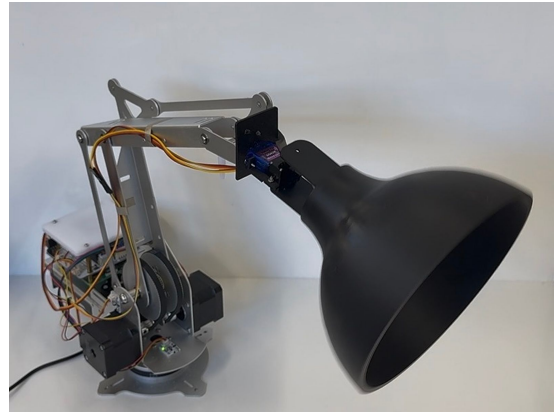
(a) HOME — lamp head pointing away from the user.



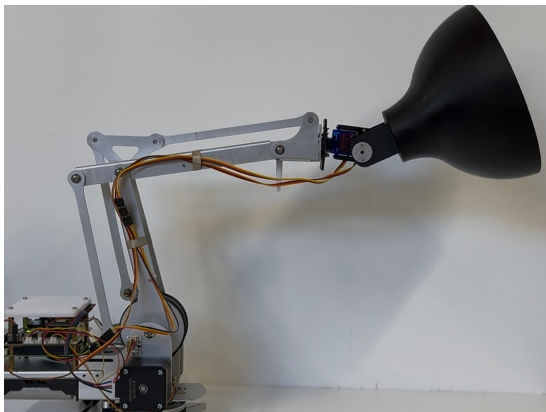
(b) LOOK — lamp head oriented toward the user.



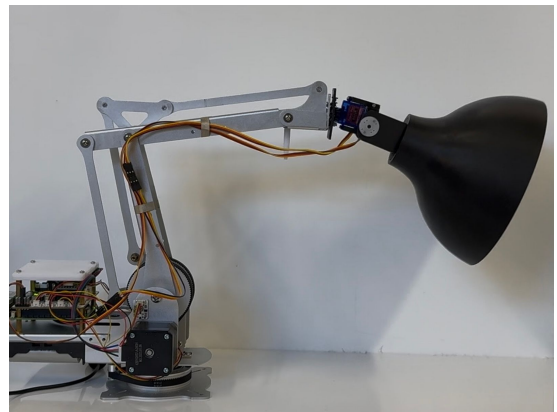
(c) SHAKE NO (left extreme).



(d) SHAKE NO (right extreme).

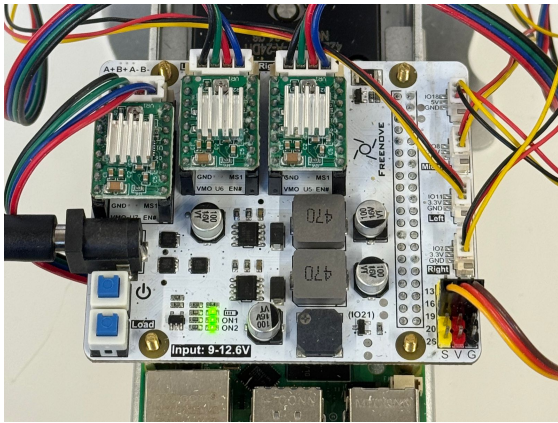


(e) NOD YES (upward pitch).

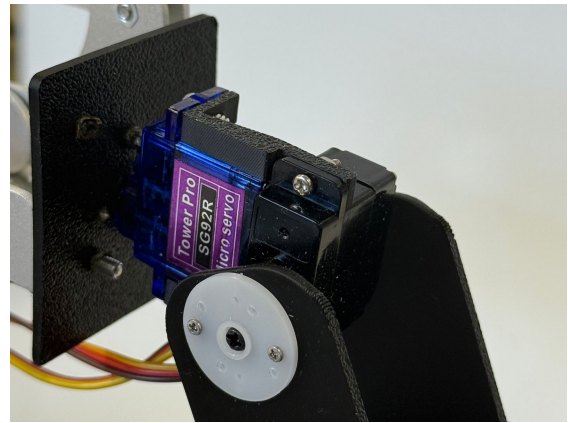


(f) NOD YES (downward pitch).

Figure 4.2: The robot’s intervention behaviours. (a–b) HOME and LOOK arm poses, shown from the side; the arm transitions between these poses to deliver the orientation cue. (c–d) SHAKE NO: a brief side-to-side head motion produced by the gimbal upon arrival at LOOK, shown from an elevated angle to make the rotation visible. (e–f) NOD YES: a small head pitch produced by the gimbal upon return to HOME. Together, these constitute the system’s complete intervention repertoire.



(a) Freenove control board with three A4988 stepper motor drivers, mounted above the Raspberry Pi 5. GPIO pins 18 and 19 route to the gimbal servos.



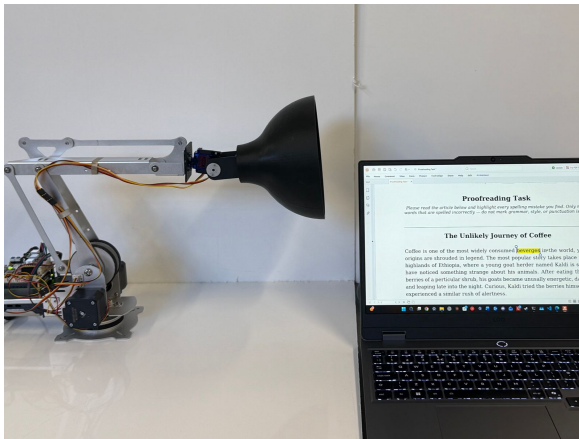
(b) Gimbal head assembly. The TowerPro SG92R micro servo provides tilt actuation; a second servo mounted orthogonally provides lateral rotation.

Figure 4.3: Hardware components. (a) Control electronics: the Freenove board handles stepper motor driving while the Raspberry Pi 5 hosts the TCP socket server. (b) Gimbal head: two SG92R micro servos enable independent (NOD YES) and (SHAKE NO) motion.

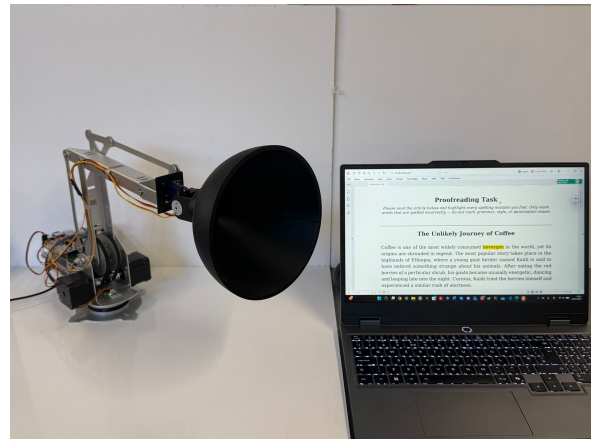
Chapter 5

User Evaluation

5.1 Aim



(a) HOME state: robot at rest during focused work.



(b) LOOK state: lamp head oriented toward the user upon distraction detection. The laptop webcam (visible at the top of the screen) serves as the sole sensing input.

Figure 5.1: The system deployed on a desk during the proofreading evaluation task. (a) Baseline HOME state during focused work. (b) LOOK state triggered upon distraction detection.

The aim of the user evaluation was to compare two conditions. The two conditions were an active, orientation-based robotic companion and a passive robot condition in which the robot did not move but was physically present. The evaluation assessed whether, they affected participants’ concentration during desk-based tasks. This evaluation addressed the research questions introduced in Chapter 1: whether the vision-based distraction detection pipeline can be used as a real-time proxy for distraction in a naturalistic single-user environment (RQ1), whether the robot’s orientation behaviour constitutes an attention cue that users can accept (RQ2), and whether the active behaviour of the robot creates a meaningful sense of co-presence similar to body doubling (RQ3).

5.2 Experimental Design

In the experimental design, there was a participant setting with two conditions. Figure 5.1 shows The deployment setup. The robot-active condition was conducted as described in Chapter 4. The robot observed the participant’s posture in real time. It responded user through the full intervention cycle in response to detected distracted state. That is, it oriented the lamp head toward the participant (HOME → LOOK), performed a brief “no” shaking motion together with this, and returned to the home position with a brief “yes” nodding motion when the participant returned to a focused state. In the robot-passive

condition, the robot was physically present in the same position on the desk. However, Robot was powered off and did not move at all during the session. It simply looked like a lamp on the desk.

I deliberately chose the passive robot condition rather than a robot-absent control condition. By maintaining the physical presence of the robot in both conditions, the difference in outcomes between the conditions is not simply whether the robot is present or absent. It depends on the robot’s movement and responsiveness. Therefore, the passive condition accounts for the novelty and visual effect of the robot’s co-presence, while in the active condition, the orientation behaviour is isolated as the active variable of interest.

To achieve balanced group sizes, the participants’ conditions were assigned using an alternating assignment method. Participants were not informed which condition was given until after the session.

5.3 Pilot Phase

Before formal data collection, I did five pilot sessions (P01–P05) to check the system functionality and identify issues that needed debugging. All pilot sessions used the robot-active condition. The pilot participants performed the proofreading task, but they did not complete the questionnaire. This is because the purpose of this stage was strictly system validation, not measurement.

Two major significant issues emerged from the pilot data. Because of this, I was able to revise the system before the main user testing. The first issue was that the robot’s response latency was measured as 5.6 seconds. Response latency means the time between distraction detection and the start of the robot’s movement. In several pilot sessions, when the robot moved to the look position, the participants had already returned to a focused state. This created a delay between the robot’s actual movement and the command, and from the participant’s perspective, the purpose of the robot’s movement became unclear. The intervention itself became ineffective. By simplifying the robot’s movement sequence, this latency was reduced to 2.0 seconds.

The second issue was that the looking-down pose feature continuously produced false-positive warnings for the system. When participants read text at the bottom of the screen or looked at the keyboard, participants who leaned their body forward repeatedly received the robot’s non-verbal cue even though they were sufficiently focused on the task. In an interview with one participant, I received feedback that they were confused because of the robot’s false-positive warning. In the main user testing, this condition itself was removed from the logic. These two changes were reflected in the v3.2 revision described in Section 4.10. It was successfully used in all sessions after P06.

5.4 Participants

The main evaluation included six participants in the robot-active condition and six participants in the robot-passive condition. Participants were recruited through convenience sampling from my personal network and members of the local community. The inclusion criteria were that participants had to be 18 years of age or older and have sufficient English reading ability to participate in the proofreading task. No diagnosis of ADHD or prior experience with ADHD was required. This is because the purpose of user testing was to assess usability across a broader population than the formative survey. I collected informations such as age range, gender, self-reported ADHD status, general focus ability, and comfort with robots through the pre-study questionnaire.

All participants signed an informed consent form before the evaluation. This study was conducted under the ethics approval stated in the Ethics Statement.

5.5 Materials

5.5.1 Proofreading Task

The task in this evaluation was the completion of a proofreading task. Participants were presented with a two page article about the unlikely journey of coffee and asked to find and highlight every spelling mistake in the article. The article contained 22 spelling errors at three different levels of difficulty. There were 7 easy spelling errors, 6 medium difficulty spelling errors, and 9 hard spelling errors. These included simple missing-letter errors, such as *unusally* instead of *unusually*. It also included more subtle phonological errors, such as *suspision* instead of *suspicion*. All errors were spelling errors only, and there were no grammatical or punctuation errors. The proofreading article and answer key were generated

with the assistance of a large language model and verified manually before use; the prompt is recorded in Appendix A.

The reasons for the choice of task are several. The task cognitively demands enough to produce variation in engagement with the task, yet is light enough to allow individuals to naturally perform the task at their desks with laptop webcams in view.

The task was not time-limited by the participant; instead, participants were told to continue until the researcher indicated the session was over. The actual stop time was determined by the researcher based on the target length of the session (seven to ten minutes) in order to create some degree of temporal uncertainty but to ensure that participants remained engaged in the session throughout its duration. Proofreading-style tasks are known to elicit reliable vigilance decrements over time, making them a suitable vehicle for studying the effects of attention-support interventions Warm et al. (2008).

5.5.2 Questionnaires

Three questionnaires were used. The pre-questionnaire administered was used to collect information regarding the demographics of the participants, their general ability to focus on tasks (on a five point scale from very poor to excellent), and their comfort with robots or AI-powered devices (on a five point scale from very uncomfortable to very comfortable). Additionally, participants were required to provide written consent to use their data from the study. I optionally recorded their video during the study.

both condition of post-questionnaire included the same questions in Section 1. Section 1 included questions regarding the participant’s perception of the length of the task (a free numeric response question that was asked before the participant was permitted to view a clock), boredom (a seven-point Likert scale question), focus during the task (a seven-point Likert scale question), the number of times they became distracted during the task (free text response question), and the source of their distractions during the task (free text response question). The question regarding the perceived length of the task was the first question on the questionnaire, and all participants were instructed not to check any clocks or devices during the task to measure the length of the task that the participants experienced — shorter was considered a deeper level of immersion into the task.

Section 2 of the questionnaire differed between conditions. For active condition, participants were asked four questions on five-point Likert scales: whether the robot helped them to remain focused on their tasks (Q6), whether the movements of the robot occurred at appropriate times (Q7), whether the movements of the robot were potentially distracting from their tasks (Q8, reverse-scored), and whether they felt comfortable working with the robot (Q9). Additionally, participants were asked how many times the robot moved during the task (Q10), how they felt when the robot moved (Q11, free text), and whether they experienced any errors in the timing of the movements of the robot (Q12). For the conditions in which the robot was passive, participants were only asked the questions regarding their comfort in the presence of the robot (Q6), whether the presence of the robot affected their focus (Q7), and how they felt while the robot was in the room (Q8, free text).

5.6 Procedure

Each session lasted approximately fifteen minutes. After obtaining signed consent from each participant and assigning them an ID number, the researcher provided each participant with a brief explanation of the experiment to be performed. The explanation included describing the reading task that would be performed, the questionnaires that would be distributed after the reading task, the use of the webcam to capture the participant’s poses for research analysis, and the fact that no videos would be saved for a result of the study; in the condition in which the robot would be active in the room, the participants were to be aware of the presence of the robot and its general functions but not of the specific behaviour that it was to exhibit during the study, to avoid priming.

The participant was seated at the workstation and the pre-study questionnaire was administered while the system was calibrating for ten seconds. During this phase, the participant was asked to sit in their usual reading posture. Following this phase, the participant was handed the task window and the following instructions were given to the participant: “Read the article and highlight every spelling mistake that you find. Only highlight words that are spelled incorrectly. Do not highlight any words for reasons other than misspelling. Continue doing task until I tell you to stop, even if you have reached the end of the article.” No time estimate was given for completing the task. Following delivery of the task, the researcher positioned themselves outside of the participant’s line-of-sight and started the session timer.

At the end of the session, the researcher said “Thank you, please stop there.” At that moment, the researcher provided the participant with the post-study questionnaire, instructing them to answer question Q1 about their perception of time before checking any clocks or phones. Following the study, the researcher conducted a debriefing session with the participant to tell the purpose of the study and the behavior of the robot.

5.7 Measures

5.7.1 Objective Measures

The pose and event data for each session was logged by the system. From these logs, three different objective measures could be extracted: the percentage of frames that contained the user focusing on the text (%focused), the percentage of frames in which the user was focusing on the text but encountering a warning (%warning), and the total number of warnings that occurred during the session. Additionally, for the sessions during which the robot was active, the number of motions that the robot made could be extracted from the logs.

The proofreading performance for each session was scored against the answer key by the researcher. Thus, for each session, the number of hits, misses, and false alarms were recorded for that session. For each session, the hit rate for the participant was calculated as the number of hits divided by 22. Additionally, the number of lines of the article that the participant had read was recorded for each session, which enabled each hit rate to be normalized to the portion of the article that the participant had completed during that session.

5.7.2 Subjective Measures

The primary measures of the participants’ subjective experience were the Likert items described above. For those who were active robot condition, items were used to measure helpfulness (Q6), timing of the requests (Q7), intrusiveness of the robot (Q8, reverse-scored), and comfort with the robot (Q9). For those who were passive condition, items measured comfort with the robot (Q6) and the effect of the robot’s presence on their focus (Q7). Both groups completed items related to the perceived duration of the task (Q1), boredom (Q2), focus during the interaction (Q3), and their estimated number of distractions during the experiment (Q4). Additionally, open-text response questions were used to gather further information from the participants, which was analyzed according to themes.

5.8 Results

The following sections present the outcomes of the twelve main-evaluation sessions (designated P06–P17) recorded under version 3.2 of the system. The results of these evaluation sessions are presented through three main sections. Sections 5.8.3–5.8.5 present the results of the behavioural data collected from system’s continuous frame and event logs; Section 5.8.6 presents the results of the post-survey of the users’ subjective opinions of the system; Section 5.8.7 presents the results of relationship between those two sets of findings; and Section 5.8.8 presents the results of the proofreading survey (which did not yield usable results). Finally, Section 5.8.9 provides a brief summary of these evaluations, which lead to the discussions presented in Chapter 6.

5.8.1 Sample, exclusions and analytic approach

Of the 12 main evaluation participants, half were assigned to each of the two conditions: robot-active (P06, P08, P10, P12, P14, P16) and robot-passive (P07, P09, P11, P13, P15, P17). These assignments were made according to the alternating assignment method described in Section 5.2. The pilot data (P01–P05) were excluded from the analysis presented in the following sections. These participants were excluded both due to the purposes for which the pilot was performed, as well as because the looking-down condition that was included in the v3.0 version of the system used during the pilot was removed in v3.2 prior to the main evaluation of the system (Section 5.3). The length of each of the recorded sessions was slightly variable between participants. The evaluation sessions were manually ended for all participants, though some participants were allowed to continue performing the task beyond the seven-minute target for the evaluation. The behavioural analyses that are presented below were all performed within a 420-second time window from the start of each participant’s assigned evaluation session.

The analytic approach was chosen with the small sample size in mind. The Mann-Whitney U test was used to determine the significance of the differences between the two conditions of each outcome; exact two-tailed p -values were calculated by enumerating the number of possible assignments of the 12 participants to the two groups ($\binom{12}{6} = 924$). For those outcomes that meet the assumptions of normality of the outcome variable (as assessed by the Shapiro-Wilk test; $p > .05$ for both groups), Welch’s t -test is also reported. For all tests, effect sizes are reported as Hedges’ g , which is a small-sample corrected measure of Cohen’s d (Hedges and Olkin (1985)). Where multiple outcomes form a pre-registered family, the Holm–Bonferroni adjustment is applied to control the family-wise error rate. One mathematical constraint of the design merits explicit acknowledgement: under exact Mann–Whitney with $N = 6$ per group, the smallest two-tailed p -value attainable is .0022, reached only when the two groups are perfectly separated. Several outcomes reported below yield $p = .065$, the value associated with the observed U statistic; given the discreteness of the test under $N = 6$ per group, multiple near-complete separations all map onto this exact p -value rather than producing a finer gradient. These outcomes should therefore be read in conjunction with their effect sizes rather than against an absolute α threshold. One additional analysis, motivated a priori by the recovery-side hypothesis derived from the body doubling literature Eagle et al. (2024), examined the count of warning episodes lasting ≥ 10 seconds. As a planned secondary analysis it is reported separately and is not included in the multiple-comparisons family.

5.8.2 System performance during the sessions

Across the twelve sessions, the vision pipeline ran at a mean of 10.4 Hz (median 10.5 Hz, range 9.8–11.1 Hz), close to the sampling target set in Section 4.10. Pose detection succeeded on essentially every frame in every session, indicating that the seated working posture and the lighting conditions of the study room were within the operating envelope of the MediaPipe pose model Lugaresi et al. (2019). No session was lost or aborted owing to camera, gimbal, or robot communication failure.

A means of evaluating the accuracy with which the system detected the five smoothed states of the participants was not possible with the data collected. While the system did record the state and feature values of each frame of the participants at a rate of approximately 10 Hz, the individual frames of each participant were not stored, preventing the assignment of ground truth labels to each frame. As such, any evaluation of detection accuracy was not possible with these data; this topic is revisited in the discussion of future work (Chapter 6).

Table 5.1: Per-participant behavioural outcomes computed over the matched 420 s task window. “Phone” counts warnings triggered by the phone-detection branch; “Long ≥ 10 s” counts warning episodes whose duration exceeded ten seconds.

ID	Condition	Focused (%)	Warnings	Phone	Long ≥ 10 s	Off-task (s)
P06	Robot-active	95.2	3	1	0	7.9
P08	Robot-active	93.3	5	2	0	17.2
P10	Robot-active	92.6	4	2	1	23.7
P12	Robot-active	91.2	7	3	1	22.5
P14	Robot-active	92.1	7	2	0	19.6
P16	Robot-active	86.9	10	3	0	38.7
P07	Robot-passive	76.5	9	3	2	81.2
P09	Robot-passive	77.9	8	6	4	76.0
P11	Robot-passive	76.8	9	6	5	84.4
P13	Robot-passive	97.5	6	5	0	6.5
P15	Robot-passive	67.3	18	10	5	112.8
P17	Robot-passive	59.8	24	20	5	155.3

5.8.3 Behavioural outcomes

The per-participant behavioural data are summarised in Table 5.1. Two patterns emerge from the individual sessions data alone, prior to any consideration of the group-level statistics: every participant who was active with the robot spent a higher proportion of their session in the focused state relative to every participant who was passive with the robot (with the single exception of participant P13, who had a 97.5% proportion of their session spent in the focused state — this figure is discussed in detail in

Table 5.2: Group-level behavioural outcomes (matched 420s window). Mean $\pm$ standard deviation reported per condition. Mann–Whitney U tests are reported with exact two-tailed p -values; p_{Holm} is the Holm-adjusted p -value across the family of seven primary outcomes (the planned secondary analysis on long warnings is excluded from this family). Welch’s t -test (p_t) is reported alongside for outcomes meeting the Shapiro–Wilk normality assumption ($p > .05$ in both groups). Hedges’ g uses the small-sample correction. ^aPlanned secondary analysis derived from the same event log; not included in the Holm correction family.

Outcome	Active (M $\pm$ SD)	Passive (M $\pm$ SD)	U	p_{MW}	p_{Holm}	p_t	g
Focused time (%)	91.9 $\pm$ 2.9	76.0 $\pm$ 12.5	32	.065	.195	.020	+1.60
Warning time (%)	5.2 $\pm$ 2.5	20.6 $\pm$ 12.0	4	.065	.195	.024	−1.68
Total off-task time (s)	21.6 $\pm$ 9.9	86.0 $\pm$ 49.8	4	.065	.195	.024	−1.68
Total warnings (n)	6.0 $\pm$ 2.5	12.3 $\pm$ 6.7	5	.093	.419	.066	−1.10
Mean warning duration (s)	3.6 $\pm$ 1.2	7.0 $\pm$ 3.5	4	.065	.195	.057	−1.26
Phone warnings (n)	2.2 $\pm$ 0.8	8.3 $\pm$ 5.9	2	.004	.032	.051	−1.30
Long warnings ≥ 10 s (n) ^a	0.33 $\pm$ 0.52	3.50 $\pm$ 2.17	4	.026	—	.010	−1.93
Looking-away warnings (n)	3.7 $\pm$ 2.1	4.0 $\pm$ 2.5	16	.937	.937	.806	−0.13

Chapter 6); and, of course, that the passive participants exhibited between three and twenty times as much off-task time as the active participants.

The group-level statistics for the eight outcome measures are presented in Table 5.2. For each of the eight outcome measures, the differences between the active and passive participants ran in the predicted direction; the effect sizes for each of the outcome measures were large. For six of the eight outcome measures, the effect sizes had $|g| \geq 1.0$ (which is generally considered to be a “very large” effect size); One outcome from the primary family survives Holm–Bonferroni correction. Thus, for example, the count of the number of phone-related warnings triggered by the participants indicates a statistically significant difference between groups ($U = 2$, $p = .004$, $p_{\text{adj}} = .032$, $g = -1.30$); the passive participants triggered nearly four times as many warnings related to the use of their phones as did the active participants with the robot. Furthermore, a planned secondary analysis of the warnings that were of a length of 10 seconds or longer (which were used to indicate the number of warnings that the participants experienced while attempting to regain their attention on the task) was also statistically significant ($U = 4$, $p = .026$, $g = -1.93$); this analysis is presented in detail in Section 5.8.4.

A further four outcomes — focused time, warning time, total off-task time, and mean warning duration — returned a p -value of .065 from the exact Mann–Whitney test. The p -value is actually the lower-bound estimate of the p -value rather than weak separation between group. The two effect sizes for each of these variables were all in the “very large” range on the Cohen’s g scale ($|g| > 1.2$), and Welch’s t -test (Table 5.2, column p_t) returned p -values that were all less than .03 except for long warnings, which were $p = .010$. The two lower results were total warnings, for which g was -1.10 and $p = .093$, and looking-away warnings, for which there was no difference between the conditions at all ($g = -0.13$, $p = .937$). The asymmetry between the number of phone warnings and looking-away warnings is discussed in Section 5.8.5.

Figure 5.2 displays the distributions of each of the six outcomes for the robot-active and robot-passive conditions. Each of the distributions for the robot-active condition is both shifted and more narrow than the distributions of the robot-passive groups; the difference in outcomes is not driven by a few participants who unusually responsive, but uniform of active condition ones. However, the dispersion of the robot-passive group indicates more in outcome between participants; while most of them exhibited outcomes similar to the mean for the robot-passive group, two participants (P15, P17) spent more than a quarter of their tasks in the warning state. These outcomes are discussed in Chapter 6.

Figure 5.3 displays the breath of the effects for the eight outcomes, ordered by the absolute effect sizes. Every outcome with a non-trivial effect ($|g| > 0.5$) favours the robot-active condition; the only outcome with a near-zero effect is the count of looking-away warnings.

5.8.4 Recovery dynamics: how long distractions persisted

Mean warning duration is the average length of time that each participant was off-task lasted before they were returned to the task by the system. This is the measure of interest for examining the proposed mechanism of body doubling; namely, whether the co-presence of robot will prevent participants from drifting away from the task, or whether it will only shorten the length of time that they are away from the task once they have drifted. Robot-active condition participants recovered from distraction to focus in 3.6 seconds on average, whereas the robot-passive condition took 7.0 seconds on average. The effect

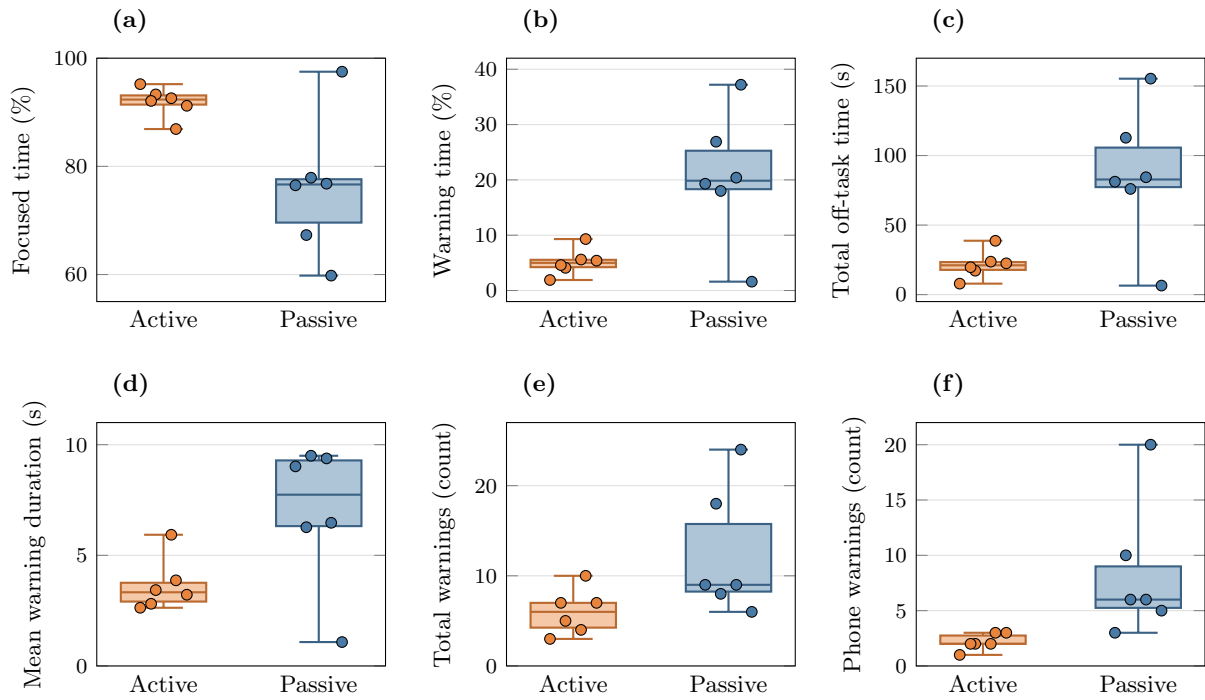


Figure 5.2: Behavioural outcomes by condition (matched 7-minute window, $N = 6$ per group). Box plots show the interquartile range and median; individual participants are overlaid as points (jittered horizontally for legibility). (a) Focused time; (b) Warning time; (c) Total off-task time; (d) Mean warning duration; (e) Total warnings; (f) Phone warnings.

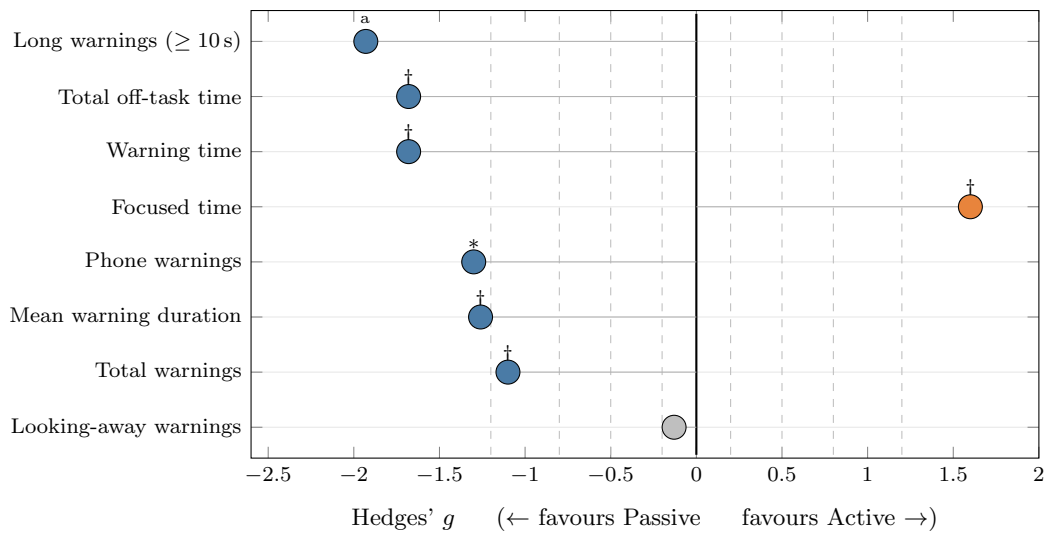


Figure 5.3: Effect sizes (Hedges' g) across the eight behavioural outcomes, ordered by magnitude. Positive values denote higher values in the robot-active condition. Significance markers (Holm-adjusted): † $p < .10$, * $p < .05$. Vertical dashed reference lines mark conventional thresholds at $|g| = 0.2, 0.5, 0.8, 1.2$. ^aPlanned secondary analysis; raw $p < .05$ but not included in the Holm correction family.

size was very large in magnitude ($g = -1.26$).

Such an aggregate statistic shows a more pointed statistic in the individual events. During the task window, robot-active participants experienced 36 warning events, while the robot-passive ones experienced 74 warning events. The distribution of these individual warning events is displayed in Figure 5.4. The robot-active experienced warning events that were narrowed down below 5 seconds in duration, and that never reached a duration of 16 seconds. In contrast, the robot-passive experienced eight warning events that were 20 seconds in duration, and the longest warning event for these lasted 43 seconds.

To make this contrast quantitative in the participant level, the count of warnings which durations are longer than 10 seconds was treated as a derived outcome (“long warnings”). Of the six robot active, four experienced no long warnings at all, and the other two experienced one long warning each. By contrast, each of the robot passive except for P13 experienced between two and five long warnings. This difference was statistically significant even after correction for multiple comparisons ($U = 4$, $p = .026$, $g = -1.93$; planned secondary analysis), and this effect size was one of the largest within the study. Thus, Both with the similar rates of looking away from the robot (Section 5.8.5), these results indicate that the robot’s main contribution to the study was on the recovery side of the attention paradigm, rather than on the side of preventing people from drifting away from the robot altogether.

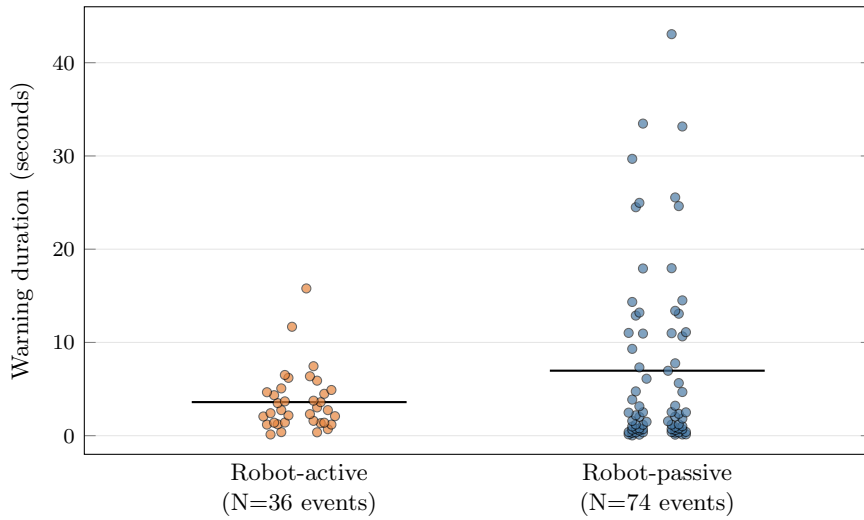


Figure 5.4: Distribution of individual warning event durations across the matched window, pooled within condition. Each point is one warning episode (robot-active: 36 events; robot-passive: 74 events). Black bars indicate the within-condition mean ($M_{\text{active}} = 3.60$ s, $M_{\text{passive}} = 6.97$ s).

5.8.5 Phone-specific behaviour and the looking-away null

These two warning categories reveal further detail about what the robot did and did not do. Looking-away warnings occurred at nearly the same rates within each of the conditions ($M = 3.7$ versus $M = 4.0$ per participant; $g = -0.13$, $p = .937$). However, phone-related warnings showed 3.8-fold difference between the two groups of participants; the robot-active participants triggered 2.2 warnings with their phone on average, yet the robot-passive participants triggered 8.3 ($g = -1.30$, $p_{\text{adj}} = .032$). Thus, the robot-active participants prompted 13 phone warnings throughout the matched time period, while the robot-passive participants prompted 50.

This interpretation is consistent with two interpretation, neither of which is mutually exclusive from the other. On the one hand, shifts in gaze away from the page (even for brief periods) are normal during the reading text, and thus may be considered inappropriate for any system to suppress this; thus, the fact that gaze shifts were not suppressed by the intervention of the robot is another desirable result. On the other hand, however, shifts in gaze to the phone indicate that the user is reading the phone rather than gaze shifts for a moment — it is only when provided with an external cue (the robot’s actions) that the user is able to focus on again

The qualitative pattern of phone warning durations supports this second interpretation. P06 (robot-active) registered a single phone warning of 1.6 seconds. P07 (robot-passive), under the same task design except for the absence of robotic feedback, registered three phone warnings, the longest of which lasted 33 seconds. The most extreme case in the sample, P17 (robot-passive), triggered twenty separate phone

warnings during the seven-minute task, four of which lasted longer than 20 seconds and one of which lasted 43 seconds. In situations where the robot’s orientation cue indicating distraction did not interrupt the user’s phone use, phone use repeatedly restarted and persisted throughout the entire session. The robot’s cue did not create a difference in the frequency of simple looking away, but it produced a strong effect on phone-related warnings. This pattern is theoretically useful. It limits the purpose of the robot to attention recovery rather than general suppression or surveillance. It also rules out the simple “observer effect” explanation that the robot’s gaze would have suppressed all forms of distraction equally.

5.8.6 Subjective experience and perception of the robot

All twelve participants completed the post-study questionnaire. The four items administered to both conditions are summarised in Figure 5.5.

Participants who were active condition felt that it was less boring than those who were passive with the robot ($M = 2.83$ versus $M = 5.00$ on a seven-point scale; $U = 2$, $p = .009$, $g = -1.92$). Furthermore, active participants felt that they were more focused on the task than passive participants ($M = 5.33$ versus $M = 4.50$ on a seven-point scale; $g = +0.63$, non-significant). Moreover, active participants tended to underestimate the length of the task; active participants felt that the task of seven minutes lasted an average of 5.83 minutes (1.17 minutes shorter than actual length), while passive participants with the robot felt that the same task lasted an average of 6.67 minutes (0.33 minutes shorter than actual length; $g = -0.36$, non-significant). Time underestimation of this kind has been reported as a behavioural correlate of cognitive engagement and flow in attention-demanding work Csikszentmihalyi (1990); This finding is in accordance with the findings regarding active versus passive participants’ feelings of focus on the task as discussed in Section 5.8.3. Reported comfort with the robot was high in both conditions ($M = 4.50$ and $M = 4.33$ respectively), suggesting that the robot’s presence was not perceived as intrusive even when it actively responded to the participant’s behaviour.

Within the robot-active condition, Likert ratings of the robot were positive on every questions (Figure 5.6). All six participants agreed or strongly agreed that the robot helped them stay focused ($M = 4.5/5$), and five of the six described its movement timing as appropriate ($M = 4.17/5$). Disagreement with the statement “the robot’s movements were distracting rather than helpful” was strong ($M = 1.83/5$): four participants gave the lowest possible rating, one gave a 2, and one (P08) gave a neutral 3. Free-text responses elaborated this pattern. P06 described the robot as “really helpful for pull me back from other things” and as “a lovely friend [seated] beside me”. P14 reported being “slightly surprised but it get me to focus again”. P12 wrote “I think it moved exact timing”, attributing appropriateness specifically to the timing placement of the intervention rather than to the gesture itself.

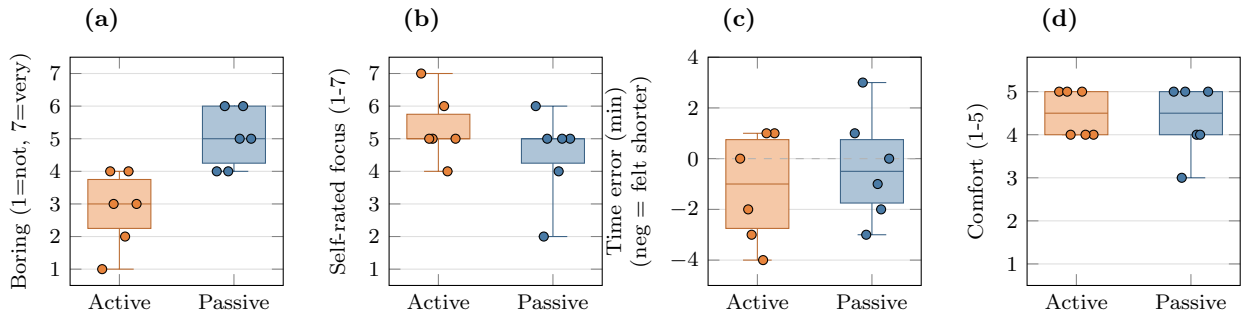


Figure 5.5: Subjective measures by condition (robot-active $N = 6$, robot-passive $N = 6$). (a) Boring rating (1 = not at all, 7 = extremely); (b) Self-rated focus (1 = not focused, 7 = very focused); (c) Time perception error in minutes (negative values indicate that the seven-minute task felt shorter than its actual duration); (d) Comfort with the robot (1 = Strongly Disagree, 5 = Strongly Agree).

5.8.7 Convergent validity and an unexpected metacognitive observation

The behavioural and subjective focus measures were derived independently of one another. The focused time measure was collected independently of the subjective measure by reviewing the system logs to determine in what fraction of the total trial the participant’s system state was in the focused state; the subjective focus measure was a single question on a Likert scale provided to each participant after the experiment. For each of the twelve participants, the two focus measures are correlated with one another

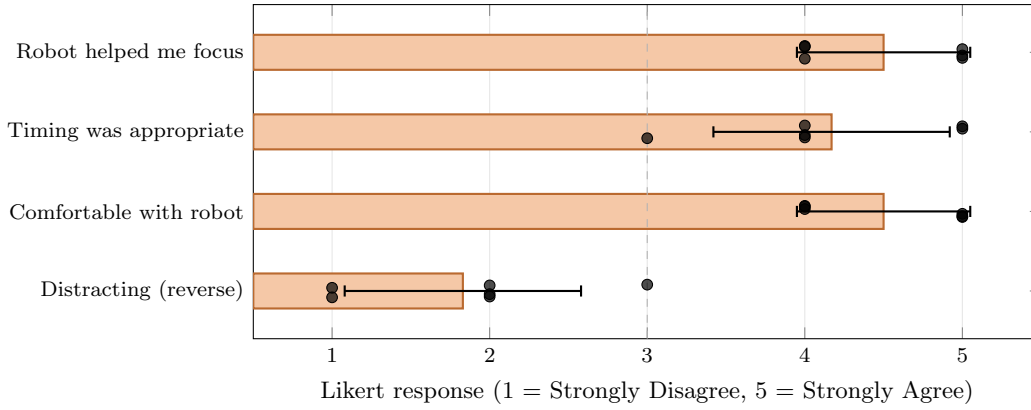


Figure 5.6: Within-condition Likert ratings of the robot (robot-active group, $N = 6$). Bars show the group mean; black horizontal lines show ± 1 standard deviation; black points show individual responses (jittered vertically for legibility). The vertical dashed line marks the neutral midpoint (3 out of 5).

(Pearson $r = .47$, $p = .12$, two-tailed). While these results are not statistically significant, they do suggest a relationship between the behavioural and subjective measures; the behavioural index reflects an element that the participants themselves recognised as their own engagement during the task, rather than arbitrary postural noise. A larger sample size would be required to test the validity of these two measures formally.

The second observation, which was not expected at the original design stage, emerged from the comparison between self-reported distraction counts (post-study Q4) and the system’s warning counts. Robot-active participants reported an average of 4.17 distractions, whereas the system recorded 6.0 warnings. This is a difference of approximately 1.4 times. Robot-passive participants reported an average of 6.17 distractions, whereas the system recorded 12.3 warnings. This is a difference of approximately 2.0 times. Both groups reported lower numbers than the system, but the gap was much larger in the robot-passive group. The gap was even clearer at the individual level. P15 reported five distractions, but the actual number of warnings was 18, which was 3.6 times higher. Passive participants tended to be inaccurate in noticing how distracted they had been. In this sense, the robot’s external feedback may have performed a metacognitive function in addition to a corrective function. In other words, it not only helped the user recover from attentional drift, but also made the attentional drift itself visible to the user. The use of external cues to support metacognitive monitoring of attention has been described in adjacent literature Fleming and Lau (2014), and the findings of this study suggest that this mechanism may also extend to physically embodied robotic feedback. A procedure for directly testing metacognitive accuracy was not included in the present design, so this interpretation should be presented cautiously. However, this pattern is consistent enough to justify follow-up in future work.

5.8.8 Proofreading task performance: floor effect

The proofreading task described in Section 5.5 was scored against the answer key after each session. Even with the reduced difficulty of the task relative to the pilot tests, most participant found fewer than ten of the 22 pre-planted spelling errors within the seven-minute time limit for the task. Furthermore, the distributions of scores for each of the conditions were too compressed to permit meaningful comparisons between the conditions; the variance between the conditions was small relative to the variance in scores due to the overall difficulty of the task. Thus, hit rates are not reported. The implications of this floor effect on the interpretation of the results from the other conditions are discussed in Section 6 (Chapter 6). However, the results reported above are unaffected by these findings in that the scores are based upon the logged data for each participant’s poses and events during the task, not the correctness of the participants in the task.

5.8.9 Summary of findings

Three findings carry the bulk of the claims of this section. The first finding is that participants in the condition in which the robot was active spent less of the task in the warning state than did participants in the condition in which the robot was passive, with all eight outcomes in this comparison showing

the predicted effect and with large effect sizes for six of the eight measures ($|g| > 1.0$). The second finding is that the strongest individual effects of the robot were upon the outcome of warnings from the phone and warnings that lasted for 10 seconds or longer, indicating that the robot's effects were felt during the recovery phase of the attention loop rather than during the prevention of distracted behaviour. The finding of no effect upon the outcome of warnings that were based upon participants looking away from the computer screen supports this result. The third is that the behavioural pattern is mirrored in the subjective data — in significantly reduced boredom, mild time underestimation, uniformly positive ratings of the robot on a Likert scale, and a moderate positive correlation between their self-reported levels of focus and their measured levels of focus (although non-significant at this sample size). Each of these findings was based upon six participants in each condition and a single session of seven minutes each; hence, each of these results is preliminary. Generalisability and persistence of these effects in longer term are discussed in Chapter 6. The following chapter introduces the literature on the topics of body-doubling and attention, and discusses the implications of these findings for the design of physical embodied systems that aim to assist humans with their attention.

Chapter 6

Discussion and Conclusion

6.1 Discussion

The evaluation has produced three findings of substantive interest. The active condition was associated with less time spent in the warning state than was observed for the passive condition; all eight of the outcomes featured in the evaluation indicated the expected effects for the active condition for robots, and the effect sizes for those outcomes were very large ($|g| > 1.0$) for six of the outcomes. The two strongest effects for the active condition for robots were for the variables that measured the number of warnings and the length of warnings that were ≥ 10 seconds in length; in each case, the effect of the robot was on the recovery portion of the attention loop rather than preventing distraction. Finally, the behavioural pattern is mirrored in the subjective data, with a significant reduction in boredom, mild underestimation of task duration, uniformly positive Likert ratings of the robot, and a moderate significant correlation between self-reported and system-measured focus. The remainder of this section takes each research question in turn before considering an unanticipated observation regarding metacognition during the analysis of the data.

6.1.1 RQ1: Reliability of the vision pipeline

The first of the research questions posed can be answered in part. More specifically, the vision pipeline accomplished its goal of delivering reliable data under the conditions of the study room; it achieved a mean rate of 10.4 Hz (Section 5.8.2), succeeded on every frame during which it was run, and experienced no failures either the camera, the gimbal, or the robot communication (Section 5.8.2). Thus, the system was found to be reliable within the context of its intended use.

A second, more robust claim regarding the reliability of the system can not be made with the data collected. For example, none of the frames were saved, making it impossible to determine the accuracy of the system in relation to the ground truth of each frame. The looking-down feature revision required during piloting (Section 5.3) was a qualitative engineering response rather than the result of a quantitative accuracy measurement. Furthermore, the moderate positive correlation between the system's measurement of focused time ($r = .47$, $p = .12$; Section 5.8.7) and the participants' self-reported measure of their focus during the study sessions, while not reaching conventional significance at this sample size, is consistent with the system's session-level focus measurement tracking participant-recognised engagement. However, still without addressing the accuracy of the system with relation to each individual frame. Thus, while the system can be considered reliable in relation to its goal of measuring focus within the context of the study room, the accuracy of the system in relation to each individual frame remains an area for future research. Furthermore, this evaluation of the system's ability to successfully detect when a user is looking down at their desktop is comparatively weaker than results obtained by other systems that apply similar techniques; for instance, the WEDAR Liu et al. (2022) system collected and manually annotated over 900,000 frames of desktop monitoring videos to create its data set of focused and non-focused desktop views. Thus, while the evaluation of this system is largely restricted to assessing the validity of the system in relation to the goals of the study, the evaluation of the component accuracies of the system remains for future research and development.

6.1.2 RQ2: Orientation as an acceptable attention cue

RQ2 asked whether the robot’s orientation behaviour constitutes a noticeable and acceptable attention cue. The subjective data answer this clearly. All six robot-active participants agreed or strongly agreed that the robot helped them stay focused ($M = 4.5/5$), five of the six rated its movement timing as appropriate ($M = 4.17/5$), and disagreement with the statement that the robot’s movements were distracting was strong ($M = 1.83/5$, with four participants giving the lowest possible rating). Additionally, This suggests that the robot’s orientation behaviour did not undermine the baseline of comfort created by the robot in its passive state.; comfort levels were high in both the control and active conditions of the study ($M = 4.5$, $M = 4.4$, respectively).

The use of a baseline condition without the active robot indicates, therefore, that the acceptability of the behavior of a passive object that is recognized as non-human is supported by these findings. This finding is in alignment with the general observation that individuals tend to respond to non-human agents that exhibit some responsive behaviour Reeves and Nass (1996). The choice of embodiment, in the form of a lamp similar to an Anglepoise lamp rather than in the form of humanoid robots, places the system somewhere in the moderate region of the anthropomorphism continuum Mori (2012); Złotowski et al. (2015). A humanoid head capable of legible gaze in the strict sense Admoni and Scassellati (2017) would have been capable of fulfilling social engagement of mutual interaction that were not to be fulfilled by the system,

and that which the participants did not appear to require. The orientation of the head to each of the participants was enough to produce the desired effect, but the more elaborate requirements of a humanoid head may have led to discomfort caused by being stared at by the system (DR5, Chapter 3). One robot-active participant (P08) provided a neutral 3 to the “distracting” item, and some of the free-text responses to Question 12 included comments about the robot moving at moments that were not perceived as drifts by the participants. Thus, it appears that the perception of the precise timing of the robot’s movements may be the factor that most contributes to the acceptability rather than the gesture itself.

6.1.3 RQ3: Co-presence and the recovery-side mechanism

Question 3 investigated whether the behavior exhibited by the robot created a sense of co-presence of the robot, similar to the concept of body doubling. Three lines of evidence demonstrate that the answers to this question is yes, and that such co-presence elaborates specific reason in the creation of such a feeling.

The first strand is the dissociation between phone warnings and looking-away warnings. Looking-away events occurred at essentially the same rate across the conditions ($M = 3.7$ vs. $M = 4.0$ per participant; $g = -0.13$), but there was a 3.8-fold difference in the number of warnings related to the phones ($M = 2.2$ vs. $M = 8.3$; $g = -1.30$, $p_{\text{adj}} = .039$; Section 5.8.5). The fact that this pattern is theoretically useful is that it helps to explain how the robot may have worked. If the robot simply made people feel watched, then one would expect that all types of distractions would have declined in a similar way. However, the data did not reflect this. For instance, people did not exhibit a decline in their simple looking-away behaviours from the participants, but they did exhibit a decline in their phone-related distractions. Thus, the robot does not appear to have equally suppressed all forms of distractions from participants. Rather, it may have been more helpful in suppressing distractions that were already occurring and continuing to occur, like the use of phones by the participants. The behavioural distinction maps onto Reason’s Reason (1990) division between attention that are brief and unavoidable when reading, and slips that last for longer periods of time and require an external trigger to end.

The second strand of evidence is for the data on recovery times (Section 5.8.4). Robot-active warnings recovered in an average of 3.6 seconds, against 7.0 seconds for robot-passive warnings ($g = -1.26$); long warnings of ≥ 10 seconds were almost absent in the robot-active condition (mean 0.33 per participant, versus 3.50 in the robot-passive condition, $g = -1.93$). Read together with the dissociation above, this supports an interpretation that the robot’s contribution was not to the reduction of the incidence of drift events, but only to the reduction of the duration of attentional drift events. The size of this effect is important to note in terms of its implications. Robot-passive participants spent roughly four times as much off-task time as compared to robot-active participants ($M = 86.0$ vs. 21.6 seconds for the seven minute period; Section 5.8.3). This difference in off-task time is mainly indicated by the long-tail of warning episodes observed in Figure 5.4. A system that consistently prevents the long tail, even without affecting the frequency of initiation occurrence altogether, has potentially disproportionate value relative to the physical actions that it requires. The third strand of evidence for the existence of the long tail is the consistency between the behavioural data and the subjective experience of the users of those technologies.

Participants who were active reported that the task was significantly less boring than those who were passively exposed to the robot ($M = 2.83$ vs. $M = 5.00$ on a seven-point scale, $g = -1.92$, $p = .009$). Furthermore, those who were actively involved in the experiment perceived the task as lasting 1.17 minutes less than its actual duration, while those who were passively exposed to the robot exhibit time perception correctly. The time underestimation phenomenon has been described in relation to the concept of cognitive engagement and flow Csikszentmihalyi (1990). The relationship between the two indices demonstrates a statistically significant correlation ($(r = .47)$) and indicates that the effect of the robot is not limited to a single index of performance.

Using Eagle et al. (2024)’s explanation of body doubling, these results suggest that the robot contributed to body doubling in two ways: through providing passive co-presence to the participant, and through providing accountability for the participant to return to the task. In other words, while the participant felt that another entity was in the room with them, and while the movements of the robot were suggestive of the need for the participant to return to the task, the presence of the robot created the sensation of body doubling.

However, the robot did not contribute to the mutual relationship that often occurs between two humans in the context of body doubling. Human body doubling suggests that each of the individuals is aware of the other’s presence and activities, and perhaps even feels a connection to the other participant. While the robot may exhibit behaviors that are suggestive of an awareness of the participant, the robot was not a true social partner to the participant. Thus, while the robot had effects on body doubling, those effects were limited to providing a sense of presence and prompting to the participant, rather than allowing for mutual interaction between the two partners.

One participant in the robot-passive condition (P13) showed a focused proportion of 97.5%, indistinguishable from the robot-active group. Recognizing this single outlier as an exception to a rule that would otherwise demonstrate that the active condition is unnecessary is supported by the fact that the removal of P13 would actually strengthen the between-condition contrast on every measure. While the data from P13 does indicate that some users are able to maintain their attention to the task without the need for any external support, the question of whether the system will be beneficial to those who do require such support. The five remaining robot-passive participants, whose times in the warning state ranged from 67 to 155 seconds within the seven minute time window, indicate that the system is beneficial to those who require support in maintaining their attention.

6.1.4 An unexpected metacognitive observation

An observation that did not appear within the original research questions became apparent through the comparison of the counts of self-reported distractions with the warnings logged by the system (Section 5.8.7). Both groups reported essentially the same counts of distractions ($M_{\text{active}} = 4.17$, $M_{\text{passive}} = 4.00$), yet the robot-passive group generated roughly twice as many warnings as those in the robot-active group. These results suggest that robot-passive participants were not as aware of their mind-wandering as were the other participants; they experienced difficulty both in recovering from mind-wandering as well as recognizing when they had wandered at all because of no external signal. Furthermore, the orientation behaviour also served a metacognitive function beyond its role of correcting the attention drift. Thus, while the participant was not directly prompted to refocus their attention, their attention drift became perceptible to the user. The use of external cues to support metacognitive monitoring of attention is a well-established theme in the cognitive literature Flavell (1979); Fleming and Lau (2014). The results of the present study suggest that the same mechanism can be applied to a physical embodied robotic interface. This interpretation is offered cautiously because: no test of metacognitive accuracy was built into the study design, the discrepancy could in principle be due to recency effects in self-report, and because this finding is exploratory and hence requires follow-up.

6.1.5 Synthesis

Ultimately, these three research questions resolved into a more specific of what the system did than the research questions themselves had anticipated. Overall, the vision system was reliable enough at the session level to track the engagement that the participants reported, though not yet at the frame level evaluation. Orientation movement of the robot was found to be acceptable to the participants who experienced it. This acceptability was not just due to the physical presence of the robot and the comfort that the participants felt in having the robot present in the room. The turning movement itself was also found to be acceptable.

The cue was mainly effective in helping the participants to recover from becoming distracted from the reading task. The cue did not effectively prevent small glance away from the reading task, but did help to prevent longer periods of distraction from the reading task. That this recovery effect was accompanied by reduced boredom, mild time underestimation, and a possible metacognitive component suggests that the value of the system is not exhausted by the count of warnings it prevented. The limitations of this evaluation, and the implications of these findings for future designs of warning systems, are discussed next.

6.2 Limitations

There are also a few limitations to the study that should be considered before drawing conclusions from the findings.

One limitation of the study is the relatively small size of the sample. Only twelve participants were used in the study, with six participants in each of the conditions. Thus, the results of the study can only provide early evidence as to the findings, rather than conclusions regarding the population in general.

Furthermore, another limitation to the study is its use of a between-subjects design. Each of the participants were only exposed to one of the two conditions. Thus, any differences between the participants may have had an effect upon the results of the study. For instance, some participants may have had more attention, proofreading, or robotics skills than others.

A within-subjects, counterbalanced design would have provided stronger evidence for the study. In this type of design, each participant would experience both of the conditions, allowing to be compared to themselves. However, this type of design was not used for the study in that the participants may become familiar with the robot’s actions in session one; they would not experience the same level of natural behavior in session two if they were aware of what that behavior would be. It could affect the results

Participants were recruited by convenience sampling from the researcher’s personal network and members of the public, which limits demographic diversity and introduces the possibility that participants who agreed to take part were systematically more comfortable with technology than the target population. The proportion of participants with ADHD or subclinical attention difficulties — the primary intended user group — is not known in advance of data collection, as ADHD status was collected as a demographic rather than used as an inclusion criterion.

The proofreading task, while ecologically appropriate as a desk-based reading task, differs from naturalistic work in several important ways. It is a single, fixed task with no variation in difficulty or structure. While participants did use their phones during these experiments (providing the data behind the analysis of the phones in Section 5.8.5), these experiments were performed in a way that does not account for broader ways of digital distraction (browsing, multitasking across applications, video consumption).

All experiments were performed by the same researcher in controlled environment. Thus, it is possible that the researcher acted as a form of social presence for the participants. Despite researcher was out of participant’s sights, which may have led to the participants exhibiting certain behaviors. It is a confounding factor that cannot be fully disengaged from the robot’s effect in the current design.

Finally, the evaluation measured short-term task performance over a single session of seven to ten minutes. The body-doubling literature suggests that the benefits of co-presence accrue over longer periods of sustained work Eagle et al. (2024), and whether the robot’s effect on attention is durable over repeated or longer sessions is an open question not addressed by the present study.

6.3 Future Work

Several directions are suggested by the limitations and findings of this project. The first of these directions is to perform a long term evaluation of the system, during which participants would use the system during their own work sessions at their own homes. Such a study would allow for the answering of several questions that were not answered in the present evaluation. For instance, it would help to answer the question of whether the effect of the robot’s attention is lasting over time, or if only effective during the initial encounter with the robot. Second, it can help to determine whether participants become used to the orientation of the robot after repeated use of the system. Third, it can help to test the system against various forms of real world distractions beyond those that were present during the proofreading task. For instance, while the proofreading task did help to minimize for reading and phone distractions, they were not completely able to account for other forms of digital distractions from the participants’ screens at home.

A within-subjects study would also help to answer some of these questions. In this type of study, each individual would experience both the active and passive sessions with the robot. In this scenario, each individual could be compared to themselves, rather than to other individuals who participated in the study. In order to create a valid experiment of this type, it would be necessary to ensure that there is enough time between the active and passive sessions to allow for the effect of the active session to fade. This time period is known as a “washout” period. By performing such a study, for instance, it would be possible to measure any additional benefit of the active robot relative to the passive presence of the robot alone.

Finally, it would also be helpful for future studies to recruit participants that are members of the target user group for this system. For instance, adults with ADHD, or who may experience difficulties with attention. Since this system is mainly designed for individuals with attention deficits and task initiation, testing the system with these individuals will provide the most realistic evaluation of the system.

At the system level, there are a number of improvements were identified in Section 4.7 but remain to be implemented. Automatic computation of the look pose of the robot based upon the position of the user in the camera frame would make the system more practical for use in the home. Moving the YOLO object detector for detecting the user’s phone to a separate process or to the Raspberry Pi would help to reduce the impact of frame rates of detecting the phone upon the frame rate of the vision system. Finally, using a TCP socket that includes a heartbeat signal would improve the reliability of the connection between the robot and the computer.

The lamp-head embodiment used in this project was designed to occupy a moderate anthropomorphic level, but no systematic comparison was made between different embodiment designs or levels of expressiveness. Future work could also investigate whether more or less anthropomorphic designs produce different patterns of acceptance, comfort, or habituation, thus extending the research on the uncanny valley Mori (2012); Złotowski et al. (2015) to the specific context of attention support.

6.4 Conclusion

In this dissertation, I describe the entire process of building and evaluating an autonomous robotic companion that can help with maintaining attention on desk-based work. Such a project was motivated both by the body-doubling literature, which suggests that the presence of another entity can help some individuals to maintain their attention on a task, as well as due to the lack of attention-assisting tools that provide a socially present but non-intrusive cue for working from home.

The system combines two central components. The first component is a vision pipeline that uses MediaPipe pose estimation and YOLO to recognise possible distraction by using phone. The second component is a lamp-inspired robotic arm that turns towards the detected user and mimics simple gestures using its two-servo gimbal mounted on the lamp head. Unlike other systems that use audible and screen notifications, the robot uses only non-verbal cues to alert the user of potential distractions.

The design of this system was informed by a formative survey of twelve individuals who identified as having ADHD. From this survey, five requirements were gathered that informed the design of the robot. These requirements included that the robot was to be approachable, non-humanoid in form, exhibit movements, and be supportive to the users, rather than potentially pressuring or surveilling them.

A user evaluation with twelve participants compared the robot-active condition with the same robot in a passive condition. Thus, it was possible to separate the effect of the robot’s orientation behaviour from the effect of its presence alone. The evaluation collected both objective and subjective data from participants including measures of perceived helpfulness, timing appropriateness, intrusiveness, and co-presence measuring both objective session metrics as well as subjective metrics from a questionnaire administered to participants after each testing session.

The project makes three main contributions. First, it provides a working prototype of a new type of attention-support tool based on embodied robotic feedback. Second, it specifies a set of design requirements for such a tool, based on empirical measurements of human and robotic behavior. Third, it provides some preliminary evidence that the use of robots to provide orientation-based attention cues may be acceptable and even useful for supporting attention in desk-based work environments.

However, it is not a clinical evaluation of the effectiveness of the system with individuals with ADHD. The main evaluation of the system did not require the participants to have an ADHD diagnosis or to identify as having ADHD. Thus, the study can be understood as a proof-of-concept evaluation of a system informed by an understanding of ADHD, but used with a population of individuals that perform desk-based work in general. While the conclusions of the evaluation cannot be broadly generalised due to the small sample size and short evaluation of task performance, they can still be used as the foundation

for future research into robotic body doubling as a potential design direction for tools that consider the needs of individuals with ADHD.

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Appendix A

Appendix A: AI Prompts/Tools

This appendix lists the AI tools used during this project, together with representative example prompts and how the resulting outputs were incorporated into the work. All design decisions, interpretation of empirical results, code logic, system architecture, hardware debugging conclusions, and the final wording of this dissertation remain my own. AI tools were used as a coding assistant, as an aid for understanding unfamiliar concepts, and as a writing aid for L^AT_EX formatting. No AI tool was used to generate research findings, statistical results, or empirical claims; all numerical values reported in this dissertation were computed from logged session data using my own analysis scripts.

Tools Used

- **ChatGPT** (OpenAI; GPT-5.4, web interface).
- **Claude** (Anthropic; Claude Opus 4.6, web interface).

Example Prompts

Prompt: “My 3D-printed lamp head is too heavy for the arm. Slicer says 140 g. Which settings should I change first to drop the mass?”

Use: Helped me understand the trade-offs between infill pattern, wall count, and shell layers. Actual parameter choices for v3 and v4 (Section 4.7) were my own.

Prompt: “Gyroid vs grid infill — which is stiffer at lower density?”

Use: Background reading for the move from grid to gyroid in v3. Justification in Section 4.7 is in my own words.

Prompt: “TikZ figure with two grouped bar charts side by side, individual points overlaid as jittered dots — show me a starting template.”

Use: Starting template for the behavioural-outcomes figures in Chapter 5. Numerical values, axis ranges, captions, and final styling were filled in by me.

Prompt: “How do I draw a boxplot in PGFPlots from raw data points instead of pre-computed quartiles?”

Use: Used for the boxplot panels in Figure 5.2. Final styling and jitter overlay added by me.

Prompt: “How do I overlay scatter points on a boxplot in PGFPlots so the points don’t sit exactly on the centre line?”

Use: Used to add jittered participant points on top of the condition boxplots in Figures 5.2 and 5.5.

Prompt: “PGFPlots horizontal bar chart with significance markers (asterisks) next to specific bars — how?”

Use: Used for the effect-size figure (Figure 5.3) where Holm-adjusted significance markers sit alongside each bar.

Prompt: “How do I draw vertical reference lines at specific x-values across a PGFPlots axis without them appearing in the legend?”

Use: Used for the $|g| = 0.2, 0.5, 0.8, 1.2$ reference lines in Figure 5.3.

Prompt: “TikZ system architecture diagram — boxes connected by labelled arrows, grouped into two regions. Give me a clean starting template.”

Use: Starting structure for Figure 4.1 (system architecture). Component names, groupings, arrow labels, and final layout were filled in by me to match the actual system.

Prompt: “How do I make a booktabs table with grouped column headers?”

Use: Reference for formatting Tables 5.1 and 5.2. Data was produced by my own analysis script.

Prompt: “How do I align decimal points in a numeric column in a booktabs table?”

Use: Used for the $M \pm SD$ and p-value columns in Table 5.2.

Prompt: “My citations are showing up as [CITE: ...] in the PDF. What’s wrong with my natbib setup?”

Use: Debugged five placeholder citations that hadn’t resolved. Bibliography entries were written by me.

Prompt: “Suggest a colourblind-safe orange and blue hex pair that prints well in greyscale.”

Use: Colour pair used throughout the Results figures. Verified by greyscale preview.

Prompt: “How do I stop my ToC from leaving an orphan section heading at the bottom of a page?”

Use: Reference for the final formatting pass on the front matter.

Prompt: “My SG90 servo on a Pi 5 GPIO with gpiozero jitters at idle. How do I stop it?”

Use: Led to learning about software vs hardware PWM, and about `LGPIOPFactory` as the Pi 5-compatible alternative to `pigpio`. Final code uses `gpiozero` with `LGPIOPFactory` and `servo.detach()` after each move (Section 4.10).

Prompt: “Why is software PWM on Linux so unstable for servos? Can the kernel scheduler interrupt the signal?”

Use: Background for understanding why the initial `RPi.GPIO` software-PWM gimbal motion looked irregular. Move to hardware PWM was confirmed empirically.

Prompt: “Which GPIO pins on a Pi 5 support hardware PWM, and can I use two at once without conflicting with I²C or SPI?”

Use: Confirmed GPIO 13 and GPIO 19 were valid choices for the rotate and tilt servos.

Prompt: “What’s the difference between hardware PWM and software PWM in plain terms?”

Use: Conceptual background before deciding which approach to use for the gimbal servos.

Prompt: “What does PWM duty cycle mean and how does it map to servo angle?”

Use: Helped me set sensible pulse-width values for the SG90 servos in the gimbal control code.

Prompt: “`threading.Lock` vs `threading.Event` for blocking new robot commands while a movement is still running — which one?”

Use: Chose the locking primitive for the *robot busy* flag in the movement dispatcher.

Prompt: “What is a daemon thread in Python and when should I use one?”

Use: Helped me decide to run the robot movement command on a daemon thread so it terminates cleanly when the main program exits (Section 4.10).

Prompt: “Plain-language explanation of Welch’s t-test vs Student’s t-test vs Mann–Whitney U?”

Use: Background before deciding the analytic approach for Chapter 5. Decision to use Mann–Whitney as primary, Welch’s as secondary, was my own.

Prompt: “What’s a p-value, in one sentence?”

Use: Sanity-check on my own understanding before writing the analytic-approach section.

Prompt: “What does ‘effect size’ mean and why is it different from a p-value?”

Use: Background for explaining why effect sizes are reported alongside p-values throughout Chapter 5.

Prompt: “What’s the difference between parametric and non-parametric tests?”

Use: Conceptual background for justifying the use of Mann–Whitney U as the primary test given the small sample size.

Prompt: “What does ‘two-tailed’ actually mean?”

Use: Sanity-check before writing ‘exact two-tailed p-values’ in the analytic-approach paragraph.

Prompt: “Smallest p-value possible from an exact Mann–Whitney with $N = 6$ per group? Why am I getting 0.065 on so many outcomes?”

Use: Helped me understand the discreteness of Mann–Whitney at small N . Informed the wording in Section 5.8.1 about reading $p = .065$ alongside effect sizes.

Prompt: “Walk me through Holm–Bonferroni. Given raw p-values [0.004, 0.026, 0.065, 0.065, 0.065, 0.065, 0.093, 0.937], what does the adjusted sequence look like?”

Use: Verified I was applying Holm–Bonferroni correctly. Adjusted p-values in Table 5.2 were computed by my own script and cross-checked against this.

Prompt: “What is a ‘family-wise error rate’ and why do I need to correct for it?”

Use: Background for the multiple-comparisons correction paragraph in Section 5.8.1.

Prompt: “Hedges’ g vs Cohen’s d — why g for small samples? What’s the correction factor for $n = 6$?”

Use: Justified the choice of Hedges’ g throughout Chapter 5.

Prompt: “What does the Shapiro–Wilk test actually test, and why does $p > .05$ mean I can assume normality rather than the other way around?”

Use: Helped me phrase the normality-check sentence in Section 5.8.1 correctly (avoiding the common mistake of treating non-rejection as proof of normality).

Prompt: “What’s the difference between standard deviation and standard error?”

Use: Sanity-check before reporting $M \pm SD$ in Table 5.2.

Prompt: “Why is a between-subjects design weaker than within-subjects, and what is a ‘washout period’?”

Use: Used in writing the Limitations and Future Work sections (6.2, 6.3) where I discuss the trade-off and propose a within-subjects follow-up.

Prompt: “What does Pearson’s $r = .47$, $p = .12$ tell me with $N = 12$? Can I claim convergent validity?”

Use: Background for the convergent-validity paragraph in Section 5.8.7. Decision to phrase the claim cautiously (‘moderate but not statistically significant’) was my own.

Prompt: “When reporting effect sizes alongside non-significant p-values at small N , how do I frame this so reviewers don’t read it as p-hacking?”

Use: Informed the analytic-approach paragraph in Section 5.8.1 and the wording of Tables 5.1–5.2.

Prompt: “Passive participants generate twice as many warnings as active ones, but self-reported distraction counts are almost identical. Is ‘metacognitive accuracy’ the right framing for this?”

Use: Pointed me to the literature on metacognitive monitoring (Flavell 1979; Fleming & Lau 2014), which I read and cited in Section 5.8.7. Interpretation in that section is my own and presented cautiously.

Prompt: “Should a planned secondary analysis go in the same Holm correction family as the primary outcomes, or reported separately?”

Use: Confirmed that the long-warnings analysis should be reported outside the Holm family. Reflected in the footnote of Table 5.2.

Prompt: “I need a 700-word fictional article on a neutral topic for a proofreading task. Plant 22 spelling errors of three difficulty levels: 7 simple missing-letter errors, 6 medium transposed/substituted letters, 9 harder plausible non-words. No grammar or punctuation errors. Generate both a participant version (errors hidden) and an answer key version (errors marked).”

Use: Generated the proofreading article “*The Unlikely Journey of Coffee*” used in all main-evaluation sessions (Section 5.5.1) and the corresponding answer key. Both versions were reviewed by me before use. The article topic was selected to be affectively neutral and non-technical. The 22 planted errors and their classifications were verified against the answer key when scoring participant responses; no error was added or removed after generation.

Summary

AI tools were used as a learning aid, a coding assistant, and a formatting reference. They were not used to generate research findings, to perform the statistical analysis itself, to design the experiment, to interpret participant data, or to write any section of this dissertation from scratch. The hardware design, software architecture, study design, data analysis, interpretation of results, and final wording of this dissertation are my own work.

Appendix B

Appendix B: Study Materials

This appendix contains the materials used in the user evaluation described in Chapter 5: the proofreading task presented to all participants, and the blank questionnaire used to collect pre- and post-study responses. Completed individual responses are not reproduced here in order to preserve participant anonymity in line with the project’s ethics approval (Application 6683); anonymised group-level results are reported in Section 5.8.

B.1 Proofreading Task

The article shown below was presented to participants on screen during the seven-minute task window. Participants were asked to highlight every spelling error they could find. The article contains 22 planted spelling errors, and the original answer key is held with the project records.

Proofreading Task — ANSWER KEY

Total errors: 22. Errors are shown in red and numbered. Confidential — researcher use only.

The Unlikely Journey of Coffee

Coffee is one of the most widely consumed **beverages**^[1] in the world, yet its origins are shrouded in legend. The most popular story takes place in the highlands of Ethiopia, where a young goat herder named Kaldi is said to have noticed something strange about his animals. After eating the red berries of a **perticular**^[2] shrub, his goats became **unusally**^[3] energetic, dancing and leaping late into the night. Curious, Kaldi tried the berries himself and experienced a similar rush of alertness.

He took his discovery to a local **monastery**^[4], where the monks were initially skeptical. One monk even threw the berries into a fire, believing they were the work of the devil. But as the beans roasted, they released a rich aroma that filled the room. The monks quickly rescued the beans from the ashes, ground them, and mixed the powder with hot water. The resulting drink helped them stay awake during long nights of prayer, and word of this remarkable substance began to spread.

By the fifteenth century, coffee had travelled across the Red Sea to Yemen, where it was being cultivated on a larger scale. The Sufi mystics of Yemen embraced coffee for its ability to help them maintain focus during **religous**^[5] **cerimonies**^[6]. From Yemen, coffee spread rapidly to Mecca, Cairo, and eventually Istanbul, where the world's first coffeehouses opened in the mid-sixteenth **centruy**^[7]. These **establisments**^[8] became important social hubs, known in Turkish as “qahveh khaneh,” or “schools of the wise.”

When coffee finally reached Europe in the seventeenth century, it faced considerable **suspision**^[9]. Some religious leaders called it “the bitter invention of Satan,” and for a time, its fate was uncertain. Legend has it that Pope Clement VIII was asked to rule on whether coffee was **acceptable**^[10] for Christians. Before making his decision, he insisted on tasting the drink himself. He found it so **delicous**^[11] that he reportedly declared it should be baptised to make it a truly Christian beverage.

Coffeehouses soon spread across Europe, particularly in England, where they came to be known as “penny universities.” For the price of a single penny, patrons could enjoy a cup of coffee and engage in stimulating conversation with scholars, merchants, and politicians. These

establishments became so **influential**^[12] that King Charles II attempted to ban them in 1675, fearing they were breeding grounds for political dissent. The ban was lifted only eleven days later after public outcry.

The expansion of European colonial empires carried coffee to new continents. The Dutch were among the first to cultivate coffee outside its original range, establishing **plantations**^[14] in Java and Ceylon. The French later introduced coffee to the **Carribean**^[13], and from there it travelled to Central and South America. Today, Brazil is the world's largest producer of coffee, **accounting**^[15] for roughly a third of global supply.

Despite its widespread popularity, coffee cultivation faces serious challenges in the twenty-first century. Climate change is altering the conditions in the “coffee belt,” the narrow band of tropical regions where coffee plants thrive. Rising **temperatures**^[16], shifting rainfall patterns, and new diseases affect the livelihoods of **millions**^[18] of farmers who depend on coffee for their income. Scientists are working to develop more **resilient**^[17] varieties, but the future of our morning cup remains uncertain.

In the Americas, coffee took on particular cultural **significance**^[19] after the Boston Tea Party of 1773. As colonial Americans rejected British tea in favour of more patriotic alternatives, coffee consumption rose dramatically. What had once been a fashionable European import became, within a generation, a staple of everyday American life. By the nineteenth century, coffee was a cornerstone of both urban cafés and rural farmhouse breakfasts, and it remains deeply **woven**^[20] into American culture today.

The twentieth century saw coffee transformed by industrial innovation. Instant coffee, first patented in 1890 but popularised during both World Wars, brought **convenience**^[21] to millions of households. Meanwhile, the invention of the espresso machine in Italy created an entirely new drinking culture, one built around speed, strength, and sociability. More recently, the so-called “third wave” of coffee has emphasised origin, craft, and flavour, treating coffee with the same care once reserved for fine wine.

Perhaps what is most **remarkable**^[22] about coffee is its ability to bring people together. From the Ethiopian goat herder to the modern office worker, coffee has served as a social lubricant, a source of energy, and a small daily ritual that punctuates our lives. Its journey from wild shrub to global commodity is a testament to the curious nature of human beings and our willingness to embrace the unfamiliar.

Error Summary & Scoring Guide

Below is the list of all 22 planted errors with explanations. Use this to score each participant. Track: (a) Hits — correctly identified errors, (b) Misses — errors the participant did not mark, (c) False alarms — non-errors the participant incorrectly marked. Also note the last line the participant read (progress marker) so scores can be normalised to how far they got.

All 22 errors are spelling mistakes. Difficulty mix — easy: 7 · medium: 6 · hard: 9

1. **"beverages"** → "beverages" · easy · missing 'a'
2. **"perticular"** → "particular" · hard · a→e substitution
3. **"unusally"** → "unusually" · easy · missing 'l'
4. **"monastery"** → "monastery" · hard · missing 'e'
5. **"religous"** → "religious" · hard · missing 'i'
6. **"cerimonies"** → "ceremonies" · hard · e→i substitution
7. **"centruy"** → "century" · medium · transposed letters
8. **"establissements"** → "establishments" · medium · missing 'h'
9. **"suspision"** → "suspicion" · hard · c→s substitution
10. **"acceptible"** → "acceptable" · medium · wrong suffix (-able not -ible)
11. **"delicous"** → "delicious" · easy · missing 'i'
12. **"influenacial"** → "influential" · hard · t→c substitution
13. **"Carribean"** → "Caribbean" · medium · double 'r' in wrong place
14. **"plantaions"** → "plantations" · easy · missing 't'
15. **"acounting"** → "accounting" · easy · missing 'c'
16. **"tempertures"** → "temperatures" · easy · missing 'a'
17. **"resiliant"** → "resilient" · hard · e→a substitution
18. **"milions"** → "millions" · easy · missing 'l'
19. **"signifigance"** → "significance" · medium · extra 'g' inserted
20. **"woved"** → "woven" · medium · n→d substitution
21. **"convinience"** → "convenience" · hard · e→i substitution
22. **"remarkeable"** → "remarkable" · hard · extra 'e' inserted

Per-participant scoring template

Participant ID: _____ Condition: _____

Last line read (page + line): _____

Hits (out of 22): _____ Misses: _____ False alarms: _____

Time to stop (or "did not finish"): _____

Notes: _____

B.2 Blank Questionnaire

The questionnaire below was administered immediately after the proofreading task. Section 1 was identical across both conditions and collected subjective ratings of focus, boredom, and perceived task duration. Section 2 differed between the robot-active and robot-passive conditions; both versions are included.

Pre-Study Questionnaire

Please complete this before the session begins.

Focus Support Robot — User Study | ~2 min

Your responses are anonymous and used for academic research purposes only. Participation is entirely voluntary.

Section 1 — About You

Q1. Age range

☐ 18-24 ☐ 25-34 ☐ 35-44 ☐ 45+

Q2. Gender

☐ Male ☐ Female ☐ Non-binary ☐ Prefer not to say

Q3. Have you been diagnosed with ADHD or any attention-related condition?

☐ Yes (formally diagnosed) ☐ Yes (self-diagnosed) ☐ No ☐ Prefer not to say

Section 2 — Focus & Robots

Q4. How would you rate your general ability to focus on tasks?

1 = Very poor 3 = Average 5 = Excellent

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Q5. How comfortable are you with robots or AI-powered devices?

1 = Very uncomfortable 3 = Neutral 5 = Very comfortable

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Section 3 — Consent

Q6. Do you consent to your anonymised data (questionnaire responses, posture logs, task results) being used for academic research and publication?

☐ Yes, I consent ☐ No, I do not consent

Q7. Do you consent to being video-recorded during the session? The recording will be framed so that your face is not visible, and the footage will be used only for research analysis.

☐ Yes, I consent ☐ No, I do not consent

Signature: _____ Date: _____

Printed name: _____

Thank you! Please hand this to the researcher — the session will begin shortly.

Post-Study Questionnaire

Please complete this immediately after the session. There are no right or wrong answers.

Focus Support Robot — User Study | ~3 min

Please answer based on how the task actually felt. Do not check any clock or phone before answering Q1.

Section 1 — Task Experience

Q1. How long did this task feel to you? (Please give your best guess.)

_____ minutes

Q2. How boring did you find the task?

Not boring at all ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 Extremely boring

Q3. How focused did you feel during the task?

Not focused at all ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 Very focused

Q4. Roughly how many times did you notice your attention wandering or getting distracted during the task?

_____ times (approximate)

Q5. What, if anything, pulled your attention away from the task?

Section 2 — About the Robot

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Q6. The robot helped me stay focused during the task.	☐	☐	☐	☐	☐

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Q7. The robot's movements came at appropriate moments (not too early, not too late).	☐	☐	☐	☐	☐

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Q8. The robot's movements were distracting rather than helpful.	☐	☐	☐	☐	☐

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Q9. I felt comfortable working with the robot present.	☐	☐	☐	☐	☐

Q10. Roughly how many times do you think the robot moved during the task?
_____ times (approximate)

Q11. How did you feel when the robot moved? (e.g. surprised, annoyed, reassured, motivated, ignored it)

Q12. Were there any moments when the robot reacted incorrectly — too early, too late, or when you were still focused?

Section 3 — Final

Q13. Any suggestions to improve the robot or this experience?

Thank you so much for your time and feedback!

Post-Study Questionnaire

Please complete this immediately after the session. There are no right or wrong answers.

Focus Support Robot — User Study | ~2 min

Please answer based on how the task actually felt. Do not check any clock or phone before answering Q1.

Section 1 — Task Experience

Q1. How long did this task feel to you? (Please give your best guess.)

_____ minutes

Q2. How boring did you find the task?

Not boring at all ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 Extremely boring

Q3. How focused did you feel during the task?

Not focused at all ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 Very focused

Q4. Roughly how many times did you notice your attention wandering or getting distracted during the task?

_____ times (approximate)

Q5. What, if anything, pulled your attention away from the task?

Section 2 — The Robot's Presence

During your task, a robot was positioned next to you but did not move or respond. Please reflect on how its presence affected you.

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Q6. I felt comfortable working with the robot present.	☐	☐	☐	☐	☐

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Q7. The robot's presence affected how I focused on the task.	☐	☐	☐	☐	☐

Q8. How would you describe the feeling of having the robot nearby? (e.g. reassuring, neutral, distracting, awkward)

Section 3 — Final

Q9. Any other comments about the task or the study?

Thank you so much for your time and feedback!